

Master's Thesis

**On the Boundaries of Communities:
Dispersal-Mediated Coexistence and Diversity
Measures in Spatial Communities**

Master's program in Evolution, Ecology and Systematics

Faculty of Biology

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Munich, May , 2026



Submitted in partial fulfillment of the requirements for the degree of M. Sc.

Abstract

Spatial species turnover is commonly attributed to environmental heterogeneity, yet dispersal and species interactions alone may also generate stable community boundaries. In this study, we investigate the formation of stable transition interfaces in homogeneous environments using generalized Lotka–Volterra reaction–diffusion models. Focusing on bistable ecological systems, we analyze traveling-wave solutions with zero propagation speed in both two-species and multispecies communities. For symmetric competitive systems, analytical approximations and numerical simulations show that the transition interface is well described by a logistic function. By transforming the dynamics into total biomass and species-contrast variables, the system can be reduced to an Allen–Cahn-like equation, yielding an approximate expression for the interface shape. Extending the model to multispecies communities reveals that stable coexistence between alternative local communities can produce persistent transition regions in which all species coexist spatially. Parameter scans further identify balance conditions between dispersal, intrinsic growth, and competition. Species with higher dispersal ability require higher growth rates and weaker competitive effects to maintain stable interfaces. Based on the emergent logistic spatial distributions, we propose a continuous formulation of β -diversity that interprets species turnover as information loss across spatial scales. Simulations suggest that fitting continuous distributions to sampled data improves the estimation of β -diversity under sufficiently dense spatial sampling. Our results demonstrate that stable ecological boundaries and spatial biodiversity patterns can emerge solely from dispersal and species interactions, even in homogeneous environments.

Zusammenfassung

Räumlicher Artenwechsel wird häufig auf Umweltheterogenität zurückgeführt; jedoch können auch allein Ausbreitung und Wechselwirkungen zwischen Arten stabile Gemeinshaftsgrenzen erzeugen. In dieser Studie untersuchen wir die Bildung stabiler Übergangsgrenzflächen in homogenen Umgebungen mithilfe verallgemeinerter Lotka–Volterra – Reaktions–Diffusions–Modelle. Mit Fokus auf bistabile ökologische Systeme analysieren wir Lösungen wandernder Wellen mit null Ausbreitungsgeschwindigkeit sowohl in Zwei–Arten – als auch in Gemeinschaften mit höherer Artanzahl. Für symmetrische Konkurrenzsysteme zeigen analytische Näherungen und numerische Simulationen, dass die Übergangsgrenzfläche gut durch eine logistische Funktion beschrieben werden kann. Durch die Transformation der Dynamik in Variablen für Gesamtbiomasse und Artenkontrast lässt sich das System auf eine Allen–Cahn – ähnliche Gleichung reduzieren, wodurch ein Näherungsausdruck für die Form der Grenzfläche gewonnen wird. Die Erweiterung des Modells auf Gemeinschaften mit höherer Artanzahl zeigt, dass stabile Koexistenz zwischen alternativen lokalen Gemeinschaften persistente Übergangsregionen erzeugen kann, in denen alle Arten räumlich koexistieren. Parameterscans identifizieren zudem Gleichgewichtsbedingungen zwischen Ausbreitung, intrinsischem Wachstum und Konkurrenz. Arten mit höherer Ausbreitungsfähigkeit benötigen höhere Wachstumsraten und schwächere Konkurrenzwirkungen, um stabile Grenzflächen aufrechtzuerhalten. Basierend auf den emergenten logistischen räumlichen Verteilungen schlagen wir eine kontinuierliche Formulierung der β -Diversität vor, die Artenwechsel als Informationsverlust über räumliche Skalen hinweg interpretiert. Simulationen legen nahe, dass die Anpassung kontinuierlicher Verteilungen an Stichprobendaten die Schätzung der β -Diversität bei hinreichend dichter räumlicher Beprobung verbessert. Unsere Ergebnisse zeigen, dass stabile ökologische Grenzen und räumliche Biodiversitätsmuster allein durch Ausbreitung und Wechselwirkungen zwischen Arten entstehen können – selbst in homogenen Umgebungen.

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1 Introduction

Understanding biodiversity is one of the central goals of modern ecology (Pimm, 1984, McCann, 2000, Sutherland et al., 2013). Explaining why so many species coexist, and how such coexistence can persist over long time scales, remains a fundamental problem in biodiversity research (May and McLean, 2007). Historically, species-rich ecosystems were often regarded as natural and self-evident, since diverse and apparently stable ecological communities are commonly observed in nature (McCann, 2000). Biodiversity was therefore traditionally viewed as an intrinsic property of ecological systems that contributes to the maintenance of ecosystem functioning and stability (Clements, 1936).

Some of earlier empirical research has further suggested that highly diverse ecosystems tend to exhibit greater functional complexity and ecological resilience (Pimentel, 1961, Hairston et al., 1968). This observation led to the classical hypothesis that biodiversity promotes ecosystem stability by increasing the capacity of communities to maintain species coexistence under environmental fluctuations and perturbations (Elton, 1977, Pimm, 1984). However, despite extensive empirical and theoretical work, the precise relationship between biodiversity and stability remains controversial (Hooper et al., 2005). One major difficulty arises from the fact that both “biodiversity” and “stability” have been defined in many different and sometimes incompatible ways (Grimm and Wissel, 1997, Domínguez-García et al., 2019), resulting in a lack of unified theoretical predictions and conclusive empirical evidence.

In contrast to purely descriptive perspectives, ecological communities can also be understood as dynamical systems governed by species interactions. With the development of complex systems theory and random matrix theory, May (1972) first proposed that large ecological systems may in fact be intrinsically difficult to stabilize when species interactions are sufficiently complex and effectively random. By linearizing the dynamics around a stationary equilibrium, the local stability of the system can be determined from the eigenvalues of the interaction matrix. Random matrix theory (Pastur, 1972, Tao et al., 2010) predicts that, for sufficiently large random systems, the eigenvalue distribution converges toward a circular region in the complex plane whose radius increases with system size. Consequently, as the number of interacting species increases, the probability that all eigenvalues remain within the stable region decreases. This result suggests that increasing complexity may destabilize ecological communities rather than stabilize them.

As a consequence, stability cannot simply be regarded as an automatic property of large ecological systems. Instead, species coexistence must be understood from the specific structural and dynamical properties that distinguish ecological communities from purely random systems. One major research direction therefore focuses on the structure of species interactions, namely the properties of the interaction matrix governing community dynamics. The type of species

interaction plays a critical role in determining stability because it constrains the sign structure of the interaction matrix. Previous studies have shown that predator–prey communities are generally more likely to remain stable than purely competitive or mutualistic systems of comparable size (Allesina and Tang, 2012). This difference arises because interaction matrices with opposite-signed symmetric elements tend to produce eigenvalue distributions confined within narrower regions of the complex plane, thereby increasing the probability of local stability. In addition to interaction type, the topological structure of ecological networks also strongly influences community stability (Bascompte and Jordano, 2007, Spiesman and Inouye, 2013, Grass et al., 2018). Network properties such as nestedness (Bascompte et al., 2003, Mariani et al., 2019), modularity (Thébault and Fontaine, 2010, Sales-Pardo, 2017), and characteristic interaction motifs (Klase and Johnson, 2017) determine the connectivity patterns and long-range organization of ecological communities (Letten and Stouffer, 2019). These structural features can substantially alter the propagation of perturbations through the network and therefore influence the persistence of species coexistence. Empirical studies further suggest that highly modular structures may promote coexistence by compartmentalizing disturbances, whereas strong nestedness is often associated with reduced dynamical stability (Bascompte and Jordano, 2007, Thébault and Fontaine, 2010).

Another important line of research focuses on the functional form of species interactions and population growth dynamics. Ecological interactions are highly nonlinear processes and may not always be adequately captured by the generalized Lotka–Volterra (GLV) framework (Case, 2000, Allesina and Tang, 2012, Barabás et al., 2018). Consequently, more realistic nonlinear functional responses, such as Holling-type interaction functions (Holling, 1959), have been incorporated into community dynamical models. Studies have suggested that these nonlinear interaction structures can substantially increase the probability of stability even in large random ecological systems (Rohr et al., 2014, Hatton et al., 2024).

However, beyond the dynamics of local communities, spatial structure also plays a fundamental role in maintaining species coexistence (Levin, 1974). Classical island biogeography theory first predicted that larger islands or habitat patches generally support more species than smaller ones (MacArthur and Wilson, 2001), a pattern that has since been confirmed across a wide range of ecosystems (Figure 1A). Building upon this observation, neutral theory explained large-scale biodiversity patterns by assuming ecological equivalence among species (Hubbell, 2001, 2005), namely that species are effectively identical in their demographic properties, dispersal abilities, and interactions. Under these assumptions, neutral models successfully reproduced several large-scale biodiversity patterns, including species abundance distributions.

More recently, theoretical studies have further demonstrated that sufficiently large spatial systems composed of many habitat patches may maintain long-term coexistence even when local exclusion occurs within individual patches (Hanski and Ovaskainen, 2000, Loreau et al.,

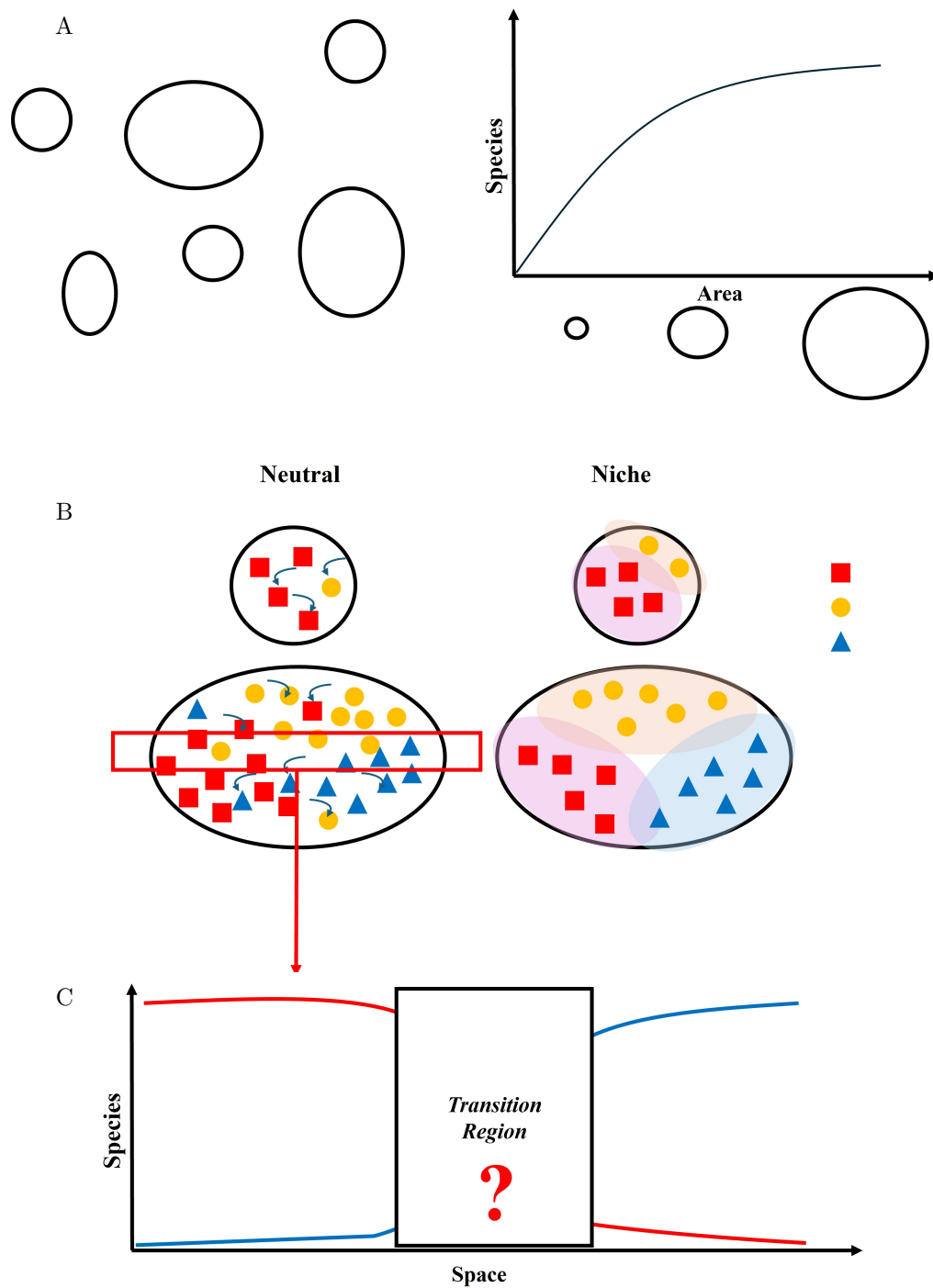


Figure 1: A. In the island biogeography, a positive relation between island area and species number is normally assumed. B. Neutral theory argues that there are more space in larger island so that more species could be maintained by only random dispersal, while niche theory argue that larger island is more probable to contain more heterogeneous habitats and thus more niche. C. A main question is, if the scenery described by neutral theory is hold, what would be the structure of the transition region between different communities, or, different species combinations.

2003, Serván et al., 2018, Miller et al., 2024) (Figure 1B). In contrast, niche theory emphasizes that larger spatial regions are more likely to contain heterogeneous environments and therefore provide a greater diversity of ecological niches (Amarasekare, 2003, Girardin, 2019, Guimarães, 2020). From this perspective, large-scale biodiversity emerges primarily through species turnover along environmental gradients. The concept of β -diversity was subsequently introduced to quantify the magnitude of such compositional turnover across space (Wang and Loreau, 2014).

Despite their differences, both neutral and niche-based perspectives ultimately highlight the importance of dispersal processes in shaping biodiversity patterns (Holmes et al., 1994, Amarasekare, 2003). Whether dispersal occurs randomly or directionally, the movement of individuals through space fundamentally links local community dynamics to regional biodiversity patterns (Gurtin, 1974, Gude et al., 2020).

Reaction–diffusion models provide a natural mathematical framework for coupling species interactions with spatial dispersal. Originally developed to explain biological pattern formation during development (Turing, 1990), reaction–diffusion systems are now widely used in mathematical biology to describe the combined effects of local interactions and spatial movement among particles, biomolecules, or ecological populations (Kondo and Miura, 2010). In community ecology, the reaction term typically represents local species interactions, whereas the diffusion term describes dispersal processes.

A central topic in reaction–diffusion theory concerns traveling wave solutions (Fife and McLeod, 1977, Gardner, 1982, Hou and Leung, 2008, Wang et al., 2024), which describe propagating interfaces connecting alternative stable states. Ecologically, these fronts may represent transitions between regions dominated by different species or communities (Figure 1C). In particular, stationary traveling waves with zero propagation speed correspond to stable spatial interfaces (Tuckwell, 1979), implying long-term species turnover without requiring explicit environmental gradients (Namba and Mimura, 1980, Carrère, 2018). Such structures therefore provide a potential mechanism for large-scale coexistence and spatial β -diversity generated purely through endogenous ecological dynamics.

The one-species case has been extensively studied and is often interpreted as the invasion or colonization of empty habitat (Holmes et al., 1994). Two-species systems have also received substantial attention because interspecific interactions can generate considerably richer spatial dynamics (Pao, 1981, Carrère, 2018, Ma et al., 2019, Wang et al., 2022). Nevertheless, despite being the simplest interacting ecological systems, the existence and stability of traveling wave solutions in competitive two-species models are still not fully understood (Hou and Leung, 2008). Extending these analyses to multispecies communities is substantially more challenging because the dimensionality and complexity of the system increase rapidly with species richness.

In this thesis, we investigate the properties of stationary traveling wave solutions with zero propagation speed in ecological reaction–diffusion systems. Using simplified two-species and multispecies models, we analyze both analytically and numerically the spatial structure of transition fronts connecting alternative community states. In particular, we focus on the conditions under which stable coexistence interfaces can emerge in the absence of explicit environmental gradients. Our results show that, under bistable interaction structures, the resulting spatial transition zones can often be approximated by logistic-like functions. Extending from the two-species case to simplified multispecies communities, we further demonstrate that dispersal and species interactions alone may generate persistent spatial turnover and heterogeneous community composition at large spatial scales. Based on these spatial population distributions, we additionally propose a continuous measure of β -diversity derived from the equilibrium abundance profiles. This framework provides a quantitative description of species turnover generated intrinsically by ecological dynamics in spatially homogeneous, or approximately neutral, environments.

2 Models and Methods

2.1 Generalized Lotka–Volterra Model

To model the dynamics of ecological communities shaped by inter-specific interactions, systems of ordinary differential equations (ODEs) are commonly employed. In such frameworks, each equation describes the temporal change in the population size of a given species. A widely used formulation is the generalized Lotka–Volterra (GLV) model (Equation 1), which can be written as

$$\frac{dN_i}{dt} = N_i \left(r_i + \sum_{j=1}^S A_{ij} N_j \right), \quad (1)$$

where S denotes the total number of species, N_i is the population size of the i th species, r_i represents its intrinsic growth rate, and A_{ij} quantifies the effect of species j on species i (Case, 2000). The interaction coefficients A_{ij} are real-valued and can encode competitive, mutualistic, or antagonistic relationships.

More elaborate models may incorporate additional ecological processes, such as environmental stochasticity or nonlinear functional responses. However, the GLV model captures key features of community dynamics, namely, intrinsic growth, effective carrying capacity, and pairwise species interactions, while remaining analytically tractable. In particular, the per-capita growth rate depends linearly on species abundances, which yields a simple form for the Jacobian matrix evaluated at equilibrium points.

Under this model, the existence and stability of a coexistence equilibrium are determined by two main conditions. First, the feasibility condition

$$\mathbf{N}^* = -\mathbf{A}^{-1} \mathbf{r} > 0 \quad (2)$$

ensures the existence of a fixed point at which all species have positive abundances (Pastur, 1972, Allesina and Tang, 2012, Rohr et al., 2014, Song and Saavedra, 2018). Second, the local stability condition

$$\max_i \operatorname{Re} [\lambda_i(\mathbf{M})] < 0, \quad (3)$$

where $\lambda_i(\mathbf{M})$ are the eigenvalues of the Jacobian matrix $\mathbf{M}$ evaluated at $\mathbf{N}^*$, guarantees that the equilibrium is locally asymptotically stable (Tao et al., 2010, Allesina and Tang, 2012). Only parameter combinations that satisfy both conditions are considered to correspond to stable ecological communities.

2.2 Diffusion-Reaction Model

In addition to species interactions, spatial community dynamics must account for the dispersal of individuals. Dispersal is commonly modeled using spatial derivatives of population density, which leads naturally to partial differential equation (PDE) formulations. A widely used approach is the diffusion model, in which dispersal is represented by a second-order spatial derivative (Källén et al., 1985). This formulation assumes that individuals move randomly and symmetrically in space. While such an assumption may allow for unrealistically rapid spread in some cases, it provides a tractable baseline description.

In reality, dispersal is influenced by multiple factors, including species interactions, environmental heterogeneity, and stochastic effects. Nevertheless, the diffusion approximation is often adopted as a null model, under the assumption that individuals disperse independently and isotropically. Alternative dispersal formulations will be discussed in subsequent sections.

Based on these considerations, reaction–diffusion models are employed to describe the spatiotemporal dynamics of biological systems. In this framework, the diffusion term represents random movement of individuals (or molecules), while the reaction term captures local growth and interactions. In developmental and cellular biology, periodic boundary conditions are often imposed, corresponding to domains that are topologically equivalent to a circle or higher-dimensional analogues (Turing, 1990). In contrast, for community ecology, we consider a finite habitat and impose no-flux (Neumann) boundary conditions (Kondo and Miura, 2010, Simpson and McCue, 2024), such that the net flux across the domain boundaries is zero.

A general form of the reaction–diffusion model is given by equation 4

$$\frac{\partial N_i}{\partial t} = D_i \frac{\partial^2 N_i}{\partial x^2} + f(N_j), \quad (4)$$

where N_i denotes the population density of the i th species, D_i is the diffusion coefficient characterizing its dispersal rate, and $f(N_j)$ represents the local reaction term as a function of all species abundances.

In this study, the generalized Lotka–Volterra model is adopted as the reaction term. The spatial domain is defined as $[-1, 1]$. The initial state is defined as two parts, where left part $[-1, 0)$ is occupied by one stable state (for example, a stable community), and right part $[0, 1]$ is occupied by the other. We analyze in detail the cases of one- and two-species systems, and further consider a representative multi-species scenario to illustrate more complex dynamical behaviors. Especially, we focus on the condition where two stable states form a stationary transition region and the property of such a transition.

2.2.1 One Species Invading Process

For the condition of a single species, the reaction–diffusion system reduces to a single partial differential equation,

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + f(u), \quad (5)$$

where $u(x, t)$ denotes the population density defined in the domain $[0, 1]$. The local population growth rate thus satisfy the condition that $f(0) = f(1) = 0$, $f(u) \geq 0$. To illustrate the evolution as well as dynamics of the front between two states, we impose a step-like initial condition,

$$u(x, 0) = \begin{cases} 1, & x < x_0, \\ 0, & x > x_0, \end{cases}$$

which represents an initially occupied region on the left invading an empty region on the right.

When $f(u) = u(1 - u)$, the model corresponds to the Fisher–KPP equation (Fisher, 1937, Simpson and McCue, 2024). The reaction term has two equilibria, $u = 0$ and $u = 1$ (Figure S.3A). Linear stability analysis shows that $u = 0$ is an unstable steady state, whereas $u = 1$ is stable. As a result, local population tend to converge to the environment capacity, once it deviated from the empty state. Consequently, the interface (or invasion front) connecting the populated state ($u = 1$) and the empty state ($u = 0$) propagates toward the unstable state.

Assuming a traveling wave solution of the form $u(x, t) = u(z)$ with $z = x - ct$ (Fife and McLeod, 1977), which indicate that the interface kept the shape and move toward the empty region, the governing equation reduces to the second-order ordinary differential equation

$$\frac{d^2 u}{dz^2} + c \frac{du}{dz} + u(1 - u) = 0.$$

The existence of traveling wave solutions requires that the wave speed satisfies $c \geq 2$ (Fisher, 1937, Breuer et al., 1994), that is, a stationary interface is impossible in Fisher-KPP model. Although closed-form solutions are generally not available (but in certain speed c , Ablowitz and Zeppetella (1979)), the leading-order approximation can be expressed as a logistic function (Figure S.3C),

$$u(z) = \frac{1}{1 + e^{z/c}} + \mathcal{O}(c^{-2}), \quad (6)$$

where $\mathcal{O}(c^{-2})$ is higher-order Infinitesimal. The logistic form approximation is already very close to the numerical solution.

To obtain a stationary front, the system must admit two (symmetrical) stable equilibria. A representative example is the Allen–Cahn model (Cahn and Hilliard, 1958) with reaction term $f(u) = u(1 - u)(u - a)$ (Simpson and McCue, 2024). The additional $(u - a)$ factor can

be interpreted as an Allee effect, reflecting the dependence of growth on population density. This system has two stable steady states, $u = 0$ and $u = 1$, and one unstable state $u = a$ (Figure S.3B). Population smaller than a cannot increase but go extinct.

The direction and speed of front propagation are determined by the associated potential function $W(u)$ of both stable states, defined via $\frac{dW}{du} = f(u)$. Evaluating the potential yields $W(0) = 0$ and $W(1) = \frac{1}{12} - \frac{a}{6}$, from which the wave speed can be expressed as $c = \sqrt{2} \left(a - \frac{1}{2}\right)$. In this formulation, the propagation dynamics are fully determined by the parameter a , and the front becomes stationary when $a = \frac{1}{2}$. In this case, the traveling wave solution admits an explicit form,

$$u(z) = \frac{1}{1 + e^{z/\sqrt{2}}}, \quad (7)$$

which corresponds exactly to a logistic profile (Figure S.3D). As a result, a symmetrical bistable reaction term will lead to a stationary interface between two stable states, making the exclusive stable state coexist.

2.2.2 Two Species Interaction–Dispersal Community

When inter-specific interactions are considered, a two-species system provides the simplest framework to capture their effects. A reaction–diffusion model with a generalized Lotka–Volterra (GLV) reaction term can be written as

$$\begin{aligned} \frac{\partial N_1}{\partial t} &= D_1 \frac{\partial^2 N_1}{\partial x^2} + N_1 (r_1 - \alpha_{1,1} N_1 - \alpha_{1,2} N_2), \\ \frac{\partial N_2}{\partial t} &= D_2 \frac{\partial^2 N_2}{\partial x^2} + N_2 (r_2 - \alpha_{2,2} N_2 - \alpha_{2,1} N_1). \end{aligned} \quad (8)$$

where D_i denotes the diffusion coefficient of species i , r_i is the intrinsic growth rate, and $\alpha_{i,j}$ represents the interaction strength of species j on species i .

The system 8 can be nondimensionalized as

$$\begin{aligned} \frac{\partial u}{\partial \tau} &= \frac{\partial^2 u}{\partial z^2} + u (1 - u - a_{1,2} v), \\ \frac{\partial v}{\partial \tau} &= D \frac{\partial^2 v}{\partial z^2} + r v (1 - v - a_{2,1} u), \end{aligned} \quad (9)$$

where the dimensionless variables and parameters are defined as

$$\begin{aligned} u &= \frac{\alpha_{1,1}}{r_1} N_1, \quad v = \frac{\alpha_{2,2}}{r_2} N_2, \quad z = \sqrt{\frac{r_1}{D_1}} x, \quad \tau = r_1 t, \\ r &= \frac{r_2}{r_1}, \quad D = \frac{D_2}{D_1}, \quad a_{1,2} = \frac{r_2 \alpha_{1,2}}{r_1 \alpha_{2,2}}, \quad a_{2,1} = \frac{r_1 \alpha_{2,1}}{r_2 \alpha_{1,1}}. \end{aligned}$$

The nondimensional system is characterized by four key parameters: D , r , $a_{1,2}$, and $a_{2,1}$, quantifying the relative dispersal ability, relative growth rate, as well as interaction strength comparable to self-regulation.

To illustrate competitive exclusion and spatial invasion dynamics, we focus on a parameter regime in which the system is locally bistable. Specifically, the intrinsic growth rates are taken to be positive, and inter-specific competition is sufficiently strong to prevent local coexistence. We first consider a symmetric case with $D = 1$, $r = 1$, and $a_{1,2} = a_{2,1}$, for which approximate analytical insights can be obtained under suitable assumptions on total population density,

$$\begin{cases} \partial_t N_1 = \nabla^2 N_1 + N_1(1 - N_1 - aN_2) \\ \partial_t N_2 = \nabla^2 N_2 + N_2(1 - N_2 - aN_1) \end{cases}. \quad (10)$$

For the general case, numerical simulations are performed across a range of parameter combinations. Specifically, we explore the effects of dispersal and growth asymmetry by varying the diffusion ratio $D \in (0, 2)$ and the relative growth rate $r \in (0, 2)$, with interval 0.02, using the model

$$\begin{aligned} \partial_t N_1 &= D \nabla^2 N_1 + r N_1(1 - N_1 - 1.5N_2) \\ \partial_t N_2 &= \nabla^2 N_2 + N_2(1 - N_2 - 1.5N_1) \end{aligned},$$

to illustrate the effect and balance between dispersal and growth in maintain the stable transition interface. In addition, we investigate the influence of interspecific competition by varying $D \in (0, 2)$ and $a_{1,2} \in (0, 2)$, interval 0.02, with the system

$$\begin{aligned} \partial_t N_1 &= D \nabla^2 N_1 + N_1(1 - N_1 - a_{1,2}N_2) \\ \partial_t N_2 &= \nabla^2 N_2 + N_2(1 - N_2 - 1.5N_1) \end{aligned}.$$

The effects of intrinsic growth rates and competition strength are analyzed separately, as these parameters primarily govern local dynamics and stability properties of the GLV system.

To quantify the stability of transition interface as well as invasion outcomes, we define the center of the interface position x^* by $u(x^*) = v(x^*)$. x^* is initially 0 since both species control one side of the original point. It is recorded at the final simulation time for each parameter combinations, which serves as an indicator of propagation speed and coexistence behavior under different parameter regimes. The transition interface is viewed as stable if x^* kept 0 during the simulation.

2.2.3 Competing Multispecies Community

More generally, for a community with multiple species, the reaction–diffusion GLV model can be written as

$$\frac{\partial N_i}{\partial t} = D_i \frac{\partial^2 N_i}{\partial x^2} + N_i \left(r_i - \sum_j A_{i,j} N_j \right), \quad (11)$$

where D_i denotes the diffusion coefficient of species i , r_i is its intrinsic growth rate, and $A_{i,j}$ quantifies the effect of species j on species i .

We focus on bistable community configurations, in which the full system does not admit a stable coexistence of all species, but instead supports two alternative stable community states. These states correspond to two subsets of species, denoted as community 1 ($C1$) and community 2 ($C2$), each of which is internally stable. As a result, the interaction between species could be divided into intra-community interactions and inter-community interactions,

$$\mathbf{A} = \begin{pmatrix} \text{Intra-}C1 & \text{Inter}_{1,2} \\ \text{Inter}_{2,1} & \text{Intra-}C2 \end{pmatrix}.$$

The dynamics can then be reformulated by partitioning species into these two groups:

$$\begin{aligned} \frac{\partial N_{1,i}}{\partial t} &= D_{1,i} \nabla^2 N_{1,i} + r_{1,i} N_{1,i} \left(1 - \sum_k A_{ik} N_{1,k} - \sum_\ell A_{i\ell} N_{2,\ell} \right), \\ \frac{\partial N_{2,j}}{\partial t} &= D_{2,j} \nabla^2 N_{2,j} + r_{2,j} N_{2,j} \left(1 - \underbrace{\sum_\ell A_{j\ell} N_{2,\ell}}_{\text{intra-community}} - \underbrace{\sum_k A_{jk} N_{1,k}}_{\text{inter-community}} \right), \end{aligned}$$

where the interaction terms are explicitly decomposed into intra-community and inter-community components.

We first analyze a simplified setting in which all species share identical diffusion coefficients, and species within the same stable subgroup are assumed to have identical parameters. Under these symmetry assumptions, analytical insights into the structure and propagation of community fronts can be obtained. More general cases, in which diffusion coefficients and interaction strengths vary across species, are investigated numerically to characterize the structure of multispecies transition fronts.

2.3 Simulation of the Partial Differential Equation Systems

For nonlinear partial differential equations such as reaction–diffusion systems, analytical solutions are generally difficult or unavailable, especially for multispecies models. To investigate the spatiotemporal dynamics and stability properties of the system, we employ finite difference methods by discretizing both space and time. Two numerical schemes are implemented: an explicit method for efficient parameter-space exploration, and a semi-implicit method for more accurate simulations in selected cases.

2.3.1 Explicit Finite Difference Method

We first apply the forward Euler method for time discretization. The spatial domain is discretized with grid spacing Δx , and time is discretized with step size Δt . Denote $N_i(x = k\Delta x, t = \ell\Delta t)$ by $u_{k,\ell}^i$. The diffusion term is approximated by the standard second-order central difference,

$$\frac{\partial^2 N_i(k\Delta x, t)}{\partial x^2} \approx \frac{u_{k+1,\ell}^i - 2u_{k,\ell}^i + u_{k-1,\ell}^i}{(\Delta x)^2}.$$

The temporal derivative is approximated by a forward difference,

$$\frac{\partial N_i(k\Delta x, \ell\Delta t)}{\partial t} \approx \frac{u_{k,\ell+1}^i - u_{k,\ell}^i}{\Delta t}.$$

Combining these approximations yields the explicit update scheme

$$u_{k,\ell+1}^i = u_{k,\ell}^i + \Delta t \left[D_i \frac{u_{k+1,\ell}^i - 2u_{k,\ell}^i + u_{k-1,\ell}^i}{(\Delta x)^2} + f_i(\mathbf{u}_{k,\ell}) \right], \quad (12)$$

where $\mathbf{u}_{k,\ell} = (u_{k,\ell}^1, \dots, u_{k,\ell}^S)$ is the vector of species abundances at grid point k and time step ℓ . This explicit scheme is computationally efficient and is therefore used for coarse simulations and parameter scans, although it is subject to stability constraints on Δt .

The transition interface center of two species case with varying parameters is done by the explicit finite difference method. With each parameter set, space $x \in [-1, 1]$ is discretized into 1000 grids, while time step size is set to $\Delta t = 1 \times 10^{-6}$, total simulation steps is 100000.

2.3.2 IMEX Semi-Implicit Method

The explicit method may become unstable when diffusion is strong or when the system exhibits stiff dynamics. To improve numerical stability and accuracy, we employ an implicit–explicit (IMEX) scheme, in which the diffusion term is treated implicitly while the reaction

term is treated explicitly. The discretized equation takes the form

$$\frac{u_{k,\ell+1}^i - u_{k,\ell}^i}{\Delta t} = D_i \nabla^2 u_{k,\ell+1}^i + f_i(\mathbf{u}_{k,\ell}). \quad (13)$$

To achieve higher spatial accuracy, the diffusion operator is approximated using a fourth-order central difference scheme,

$$\left. \frac{\partial^2 N_i}{\partial x^2} \right|_{k,\ell} \approx \frac{-u_{k+2,\ell}^i + 16u_{k+1,\ell}^i - 30u_{k,\ell}^i + 16u_{k-1,\ell}^i - u_{k-2,\ell}^i}{12(\Delta x)^2}.$$

Substituting this into the IMEX formulation (Equation 13) yields a linear system for $\mathbf{u}_{\ell+1}$ at each time step,

$$u_{k,\ell}^i + \Delta t f_i(\mathbf{u}_{k,\ell}) = u_{k,\ell+1}^i - \Delta t \left[\frac{-u_{k+2,\ell+1}^i + 16u_{k+1,\ell+1}^i - 30u_{k,\ell+1}^i + 16u_{k-1,\ell+1}^i - u_{k-2,\ell+1}^i}{12(\Delta x)^2} \right]. \quad (14)$$

In each step ℓ , the left part can be calculated. The right part is in the form of a linear combination of values in near grids in next step. The resulting system can be written in matrix form and solved efficiently using sparse linear algebra techniques.

To get a higher accuracy, we computed the error or loss of simulation steps. Change of population size at each space grid is calculated between steps, the largest of which is defined as the maxima loss. Simulation stops if maxima loss of the step is lower than a set threshold, or the number of steps reach the upper limit. Threshold of the loss is set to 1×10^{-6} for models with an approximately stable solution, and 1×10^{-8} for others. The upper limit of simulation steps is set to 1×10^7 .

Compared to the explicit scheme, the IMEX method requires solving a linear system at each time step and is therefore computationally more expensive. Consequently, it is applied only to selected cases where higher numerical stability and accuracy are required. Specifically, we simulated the symmetrical two species model where $a = 1.5$, asymmetric two species model where $D = 1.25$ and $r = 0.75$, as well as multi-specific models by the IMEX semi-implicit method.

2.4 Logistic Test for Equilibrium Distribution

The simulated spatial population profiles exhibit sigmoid-like shapes. Moreover, in the Allen–Cahn model, the traveling front admits a logistic profile under certain conditions. Motivated by this observation, we assess how well a logistic function can approximate the equi-

librium population distributions, particularly in bistable communities where spatial coexistence may arise.

For a given species with asymptotic equilibrium abundances K_1 and K_2 in two alternative community states, we consider the logistic function

$$N_L(x) = \frac{K_1 - K_2}{1 + e^{-k(x-x_0)}} + K_2, \quad (15)$$

where x_0 denotes the location of the interface (center of the front), and k characterizes the steepness of the spatial gradient (cline). This formulation ensures that $N_L(x) \rightarrow K_2$ as $x \rightarrow -\infty$ and $N_L(x) \rightarrow K_1$ as $x \rightarrow +\infty$.

In the special case where a species is present only in one community (e.g., absent in the second state), the model reduces to

$$N_L(x) = \frac{K}{1 + e^{-k(x-x_0)}}.$$

After numerical simulation, the spatial distributions of all species are assumed to have reached steady state. Each profile is then fitted to the corresponding logistic function. The fitting residual is defined as

$$s(x) = N(x) - N_L(x),$$

and the goodness of fit is quantified using the coefficient of determination R^2 for each regression.

2.5 Measure of β -Diversity

Calculate β -diversity with discrete sample and continuous distribution

Quantifying biodiversity is essential for understanding the stability and spatial organization of ecological communities. α -diversity characterizes biodiversity within a local (well-mixed or nearly homogeneous) community, whereas γ -diversity describes biodiversity at the scale of the entire system (global community). These measures depend on both species richness and relative abundances.

In this study, we employ the Shannon–Weaver index, an information-theoretic measure that quantifies the uncertainty associated with randomly sampling a species. For discrete samples,

the α - and γ -diversity are defined as

$$\begin{aligned}\alpha_k &= - \sum_i p_{k,i} \ln(p_{k,i}), & p_{k,i} &= \frac{N_{k,i}}{\sum_j N_{k,j}}, \\ \gamma &= - \sum_i q_i \ln(q_i), & q_i &= \frac{\sum_k N_{k,i}}{\sum_j \sum_k N_{k,j}}.\end{aligned}\tag{16}$$

In addition, β -diversity measures the variation in community composition across space and is often interpreted as a measure of species turnover. A common definition expresses β -diversity as the ratio between γ -diversity and the average α -diversity,

$$\beta = \frac{\gamma}{\frac{1}{n} \sum_k \alpha_k}.\tag{17}$$

When continuous spatial distributions $N_i(x)$ are available (rather than discrete samples $N_{k,i}$), these definitions can be extended accordingly. The local diversity at position x is defined as

$$\begin{aligned}\alpha(x) &= - \sum_i p_i(x) \ln(p_i(x)), & p_i(x) &= \frac{N_i(x)}{\sum_j N_j(x)}, \\ \gamma &= - \sum_i q_i \ln(q_i), & q_i &= \frac{\int N_i(x) dx}{\sum_j \int N_j(x) dx}.\end{aligned}\tag{18}$$

Accordingly, the continuous analogue of β -diversity is given by

$$\beta = \frac{\gamma}{\frac{1}{L} \int \alpha(x) dx},\tag{19}$$

where L denotes the length of the spatial domain. For simulated distribution in this work, we calculate β -diversity by equation 18 and equation 19 as the theoretical value.

Based on this framework, we propose a continuous-distribution approach for measuring β -diversity in spatial ecological systems. Given a set of spatial abundance data, the species distributions can first be approximated by logistic functions, after which biodiversity indices can be evaluated analytically from the corresponding continuous distributions.

Sampling Discrete Data from Continuous Distributions

To evaluate the proposed diversity measures under realistic sampling conditions, we generated discrete community samples from the simulated continuous population distributions. These sampled datasets were then used to estimate β -diversity and compared with the theoretical values calculated directly from the simulated distributions.

First, a set of random sampling locations ($k = 5, 10$) was selected within the simulated

spatial domain $[-1, 1]$. At each sampling location x , the probability of observing an individual from species i was assumed to be proportional to its local population abundance,

$$p_i(x) = \frac{N_i(x)}{\sum_j N_j(x)}.$$

We then sampled $n = 20, 50, 100$ individuals independently at each location according to the probability distribution $\{p_i(x)\}$. Consequently, the sampled abundances of the species, denoted by $n_i(x)$, followed a multinomial distribution,

$$P(n_1(x), n_2(x), \dots, n_S(x)) = \frac{n!}{n_1(x)! \dots n_S(x)!} \prod_{i=1}^S p_i(x)^{n_i(x)}, \quad \sum_{i=1}^S n_i(x) = n.$$

For each sampled dataset, we first calculated β -diversity directly using Equations 16 and 17, referred to here as the discrete β estimate. We then fitted the sampled abundance distributions of each species to the logistic function (Equation 15) and applied Equations 18 and 19 to obtain a continuous β estimate based on the reconstructed spatial distributions.

For each parameter combination of k and n , we generated 50 independent replicate samples. Both the discrete and continuous estimates of β -diversity were treated as samples drawn from their respective sampling distributions. The mean values obtained from the two methods were then compared with the theoretical β -diversity calculated directly from the simulated continuous distributions. Variances and mean squared errors (MSEs) were subsequently computed to evaluate and compare the accuracy of the two estimation methods.

3 Results

3.1 Equilibrium Distribution of Two Symmetric Species

We first consider the symmetric case of the two-species system, in which $D = 1$, $r = 1$, and $a_{1,2} = a_{2,1} = a$. Under these assumptions, the system can be written in the simplified non-dimensional form (Equation 10),

$$\begin{cases} \partial_t N_1 = \nabla^2 N_1 + N_1(1 - N_1 - aN_2), \\ \partial_t N_2 = \nabla^2 N_2 + N_2(1 - N_2 - aN_1). \end{cases}$$

We focus on the regime $a > 1$, for which the local dynamics are bistable and the two species cannot coexist in a homogeneous local community. Nevertheless, spatial coexistence may still emerge at the global scale because the two species are dynamically symmetric and neither has an intrinsic advantage. In this sense, the system resembles the Allen–Cahn model, as both exhibit two stable states connected by a transition front. We therefore seek an approximate analytical description of the spatial interface separating the two species. Throughout this section, we assume that species 1 occupies the left side of the domain, whereas species 2 dominates the right side.

3.1.1 Approximate solution

We conjecture that the two stable states can coexist in system 10, due to its dynamical similarity to the Allen–Cahn model. Under this assumption, the resulting stable profile should correspond to a transition from species 1 to species 2, which can equivalently be described by the transition of the population difference $v = N_1 - N_2$. Motivated by this observation, we introduce the following linear transformation as equation 20.

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} N_1 \\ N_2 \end{pmatrix}, \quad (20)$$

where the variables u and v represent the total population density and the population contrast between the two species, respectively. Substituting these transformation into the governing equations yields

$$\begin{cases} \partial_t u = \nabla_x^2 u + u - (N_1^2 + N_2^2) - 2aN_1N_2, \\ \partial_t v = \nabla_x^2 v + v - (N_1^2 - N_2^2). \end{cases}$$

There still remains the second-order term of N_1 and N_2 . To substitute these term, we also consider the combinations of second-order term of u and v ,

$$u^2 = N_1^2 + 2N_1N_2 + N_2^2, \quad v^2 = N_1^2 - 2N_1N_2 + N_2^2, \quad uv = N_1^2 - N_2^2.$$

Thus, using the identities

$$\begin{aligned} N_1^2 + N_2^2 &= \frac{u^2 + v^2}{2}, \\ N_1^2 - N_2^2 &= uv, \\ N_1N_2 &= \frac{u^2 - v^2}{4}, \end{aligned}$$

which follow directly from the linear transformation between (N_1, N_2) and (u, v) , the system can be rewritten as equation 21.

$$\begin{cases} \partial_t u = \nabla_x^2 u + u - \frac{1+a}{2}u^2 - \frac{1-a}{2}v^2, \\ \partial_t v = \nabla_x^2 v + v(1-u). \end{cases} \quad (21)$$

Since we assume that v describes the main dynamics, we now analyze the dynamics of the total density variable u first. Consider the condition where $a = 1$, two species are identical. The system would be

$$\begin{cases} \partial_t u = \nabla_x^2 u + u(1-u), \\ \partial_t v = \nabla_x^2 v + v(1-u). \end{cases}$$

Total population size follows the Fisher-KPP dynamics, thus finally be 1 anywhere. After that, the dynamics of v is just the random diffusion $\partial_t v = \nabla_x^2 v$, also get to uniform everywhere. In this trivial case, since the inter-specific and intra-specific competitions are identical, the local carrying capacity is always fully occupied. Consequently, the total population size remains constant. When it comes to our case, $a > 1$, the total population size can not maintain, because local competition always decrease it. However, considering that competition is only high in the middle where both species is not dominant, we could still assume that $u(x) = 1 - \varphi(x)$, where $\varphi(x)$ is comparable small. That is, biologically, two species nearly occupied all the capacity, or at least have the potential to increase to the capacity. Under the assumption that the total population density varies weakly in space and time, we approximate

$$\partial_t u \approx 0, \quad \nabla_x^2 u \approx 0.$$

The equation for u then reduces to the algebraic relation

$$\frac{1+a}{2}u^2 - u + \frac{1-a}{2}v^2 = 0.$$

The algebraic relation for u can be solved explicitly in terms of v , as $u = \frac{1 \pm \sqrt{1 + (a^2 - 1)v^2}}{1+a}$. Since $u = N_1 + N_2 \geq 0$, only the positive branch is biologically meaningful, yielding

$$u = \frac{1 + \sqrt{1 + (a^2 - 1)v^2}}{1+a}. \quad (22)$$

Substituting expression 22 into equation 21 for v gives

$$\partial_t v = \nabla^2 v + v \left(1 - \frac{1 + \sqrt{1 + (a^2 - 1)v^2}}{1+a} \right). \quad (23)$$

The effective reaction term $f(v) = v \left(1 - \frac{1 + \sqrt{1 + (a^2 - 1)v^2}}{1+a} \right)$ has three equilibria located at $v = 0, \pm 1$, and is antisymmetric with respect to the origin. The qualitative shape of this function resembles the Allen–Cahn reaction term $g(v) = rv(1 - v^2)$.

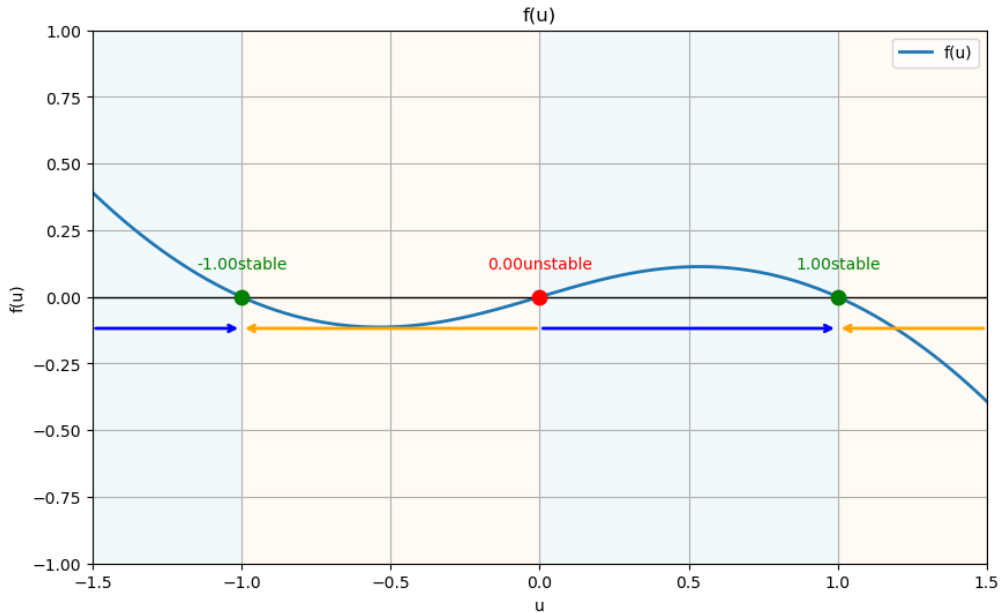


Figure 2: The graph of $f(v)$ for the simplified two symmetric species model. The curve is similar to what is in the Allen–Cahn model, where there are two stable states in both side and an instable one in the middle.

To derive an effective approximation, we expand $f(v)/v$ around $v = 0$:

$$\begin{aligned} \frac{f(v)}{v} &= 1 - \frac{1 + \sqrt{1 + (a^2 - 1)v^2}}{1 + a} \\ &= \frac{a}{1 + a} - \frac{1}{1 + a} \left(1 + \frac{(a^2 - 1)v^2}{2} + \mathcal{O}(v^4) \right) \\ &\approx \frac{a - 1}{a + 1} - \frac{a - 1}{2} v^2. \end{aligned}$$

This expansion suggests that the leading-order dynamics near $v = 0$ can be approximated by an Allen–Cahn-type equation with effective coefficient

$$r \approx \frac{a - 1}{a + 1}.$$

This approximation is further supported by the local slope of the reaction term at the origin. Differentiating $f(v)$ yields

$$f'(v) = v(1 - a)v(1 + (a^2 - 1)v^2)^{-\frac{1}{2}} + \left(1 - \frac{1 + \sqrt{1 + (a^2 - 1)v^2}}{1 + a} \right).$$

Evaluating at $v = 0$ gives

$$f'(0) = \frac{a - 1}{a + 1}.$$

Similarly, for the Allen–Cahn reaction term $g(v) = rv(1 - v^2)$, we obtain

$$g'(v) = r(1 - 3v^2), \quad g'(0) = r.$$

Matching the slopes at the origin therefore leads to the approximation

$$f(v) \approx rv(1 - v^2), \quad r = \frac{a - 1}{a + 1}. \quad (24)$$

Consequently, the interaction front can be approximated by the Allen–Cahn traveling-wave profile (keep $v(-\infty) = -1$, $v(\infty) = 1$),

$$v(z) \approx \frac{-2}{1 + e^{\sqrt{2r}z}} + 1, \quad (25)$$

following which, considering $u \approx 1$ we also get,

$$\begin{cases} N_1 = \frac{u+v}{2} = \frac{1}{1+e^{-\sqrt{2r}z}} \\ N_1 = \frac{u-v}{2} = \frac{1}{1+e^{\sqrt{2r}z}} \end{cases}.$$

3.1.2 Numerical solution

We next performed numerical simulations of the symmetric system with interaction strength $a = 1.5$. The initial condition was chosen such that species 1 fully occupied the left half of the domain ($x \in (-1, 0)$), whereas species 2 occupied the right half ($x \in (0, 1)$). Starting from this sharp interface, a transition region formed near the center of the domain and gradually relaxed toward a stationary profile. The spatial distributions of both species exhibited sigmoid-like shapes and remained symmetric with respect to $x = 0$ throughout the simulation (Figure 3A).

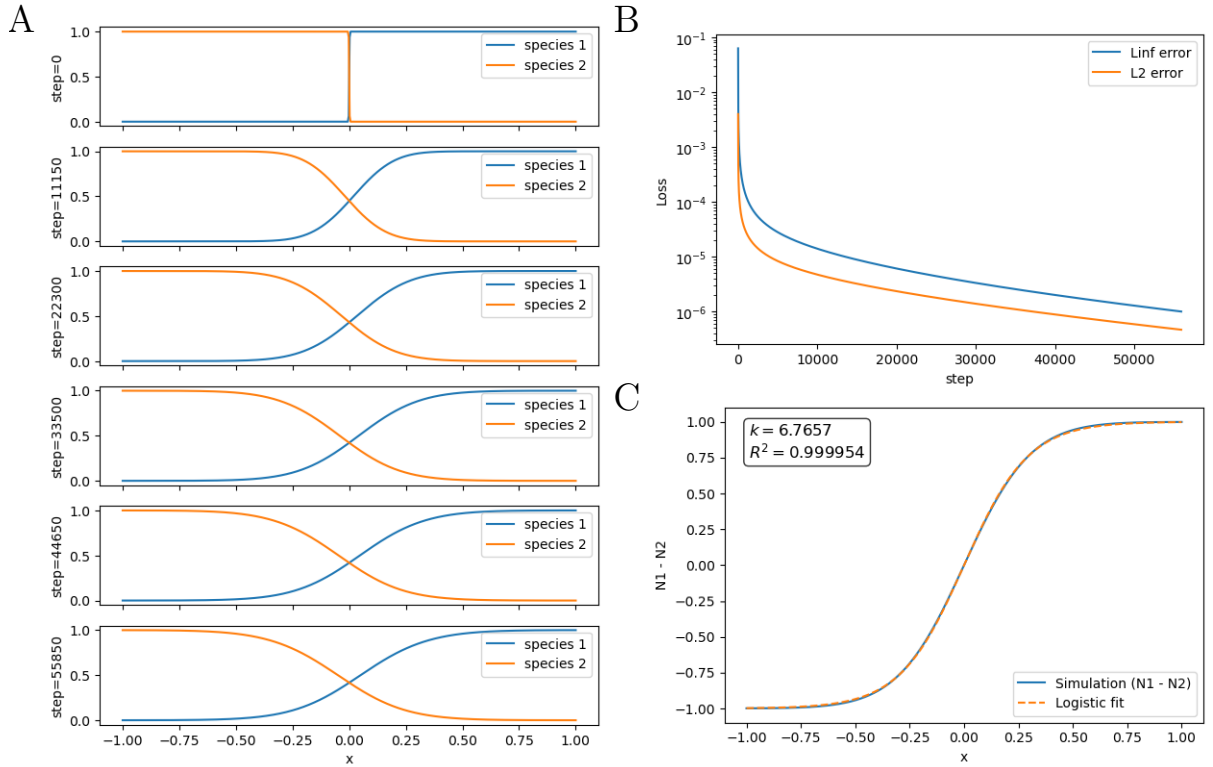


Figure 3: (A) Spatial dynamics of two symmetric competing species at simulation steps 0, 11150, 22300, 33500, 44650, and 55850. A smooth transition region emerges from the initially sharp boundary and remains stable throughout the simulation. (B) Convergence error, defined as the maximum difference between consecutive simulation steps. The simulation terminates when the error falls below 10^{-6} . (C) Logistic regression of the final profile of $N_1 - N_2$, yielding $k = 0.67657$ and $R^2 > 1 - 10^{-4}$.

To quantify convergence, we measured the maximum absolute difference between consecutive simulation steps. The loss decreased approximately exponentially over time and continued to decay until the system reached equilibrium, indicating the stability of the transition front (Figure 3B). The final distribution of $N_1 - N_2$ (obtained when the maximum stepwise error was below 10^{-6}) was fitted to a logistic function with fixed asymptotic values $K_1 = 1$, $K_2 = -1$, and center $x_0 = 0$ (Figure 3C). The logistic approximation showed excellent agreement with the simulated profile. The fitted central slope was $k = 0.676$ (after rescaling), which is close to the

theoretical prediction

$$\hat{k} = \sqrt{2r} = 0.632.$$

To further examine the analytical approximation, we analyzed the transformed variables $u = N_1 + N_2$, $v = N_1 - N_2$. In the theoretical derivation, the total density u was assumed to remain approximately constant and close to 1. This assumption is supported by the simulations: away from the interface, where only one species dominates, the total density remained close to the carrying capacity. Near the transition region, however, u was slightly lower than 1, reflecting the reduction in total population density caused by strong inter-specific competition (Figure 4B). As the simulation evolved, the deviation of u from unity became increasingly localized around the interface.

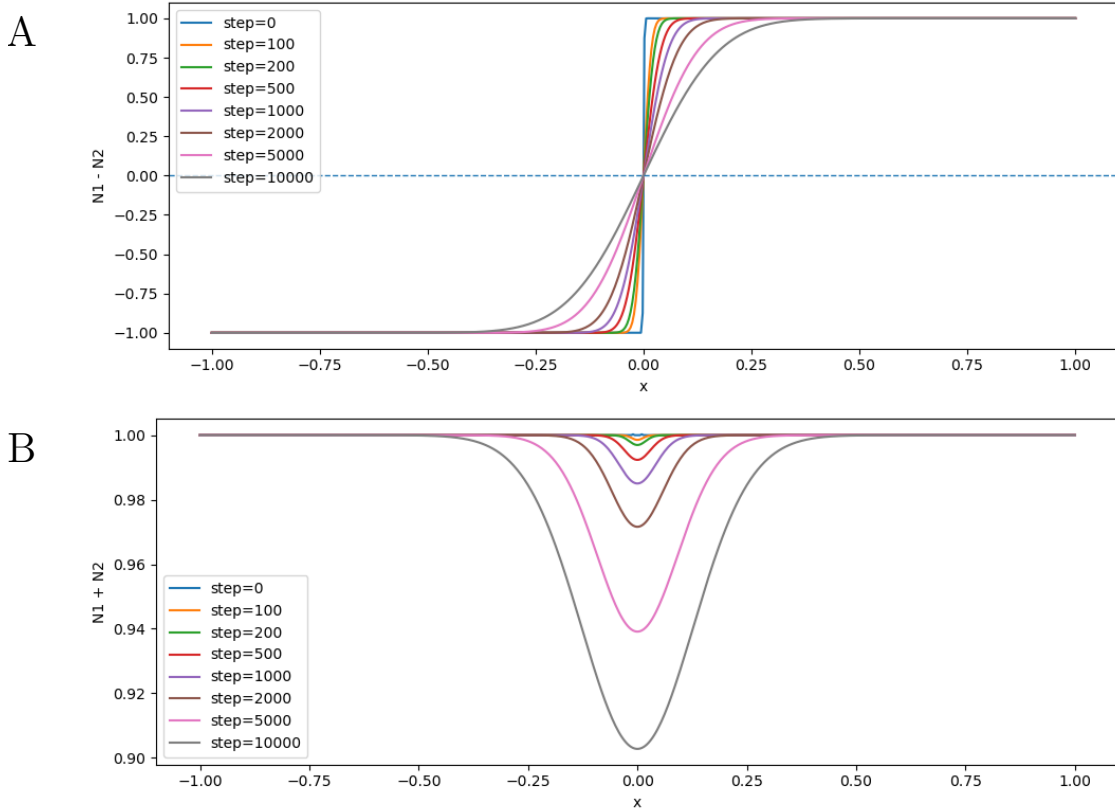


Figure 4: Spatial distributions of $v = N_1 - N_2$ and $u = N_1 + N_2$ during the simulation. Curves correspond to simulation steps 0, 100, 200, 500, 1000, 2000, 5000, and 10000. At later times, the profiles converge toward stationary distributions and therefore become visually indistinguishable.

In contrast, the difference variable v evolved from the initial step-like profile into a smooth sigmoid-shaped distribution (Figure 4A), consistent with the predicted logistic traveling-wave form. It kept cross $(0, 0)$, indicates that the dynamics is maintained symmetric during the simulation.

3.1.3 Transition slope

To evaluate the validity of the analytical approximation over different interaction strengths, we further simulated the symmetric system for varying values of

$$q = a - 1,$$

where a denotes the interspecific competition coefficient. And measured the central slope k to compare with predicting $\hat{k}$. As q increased, the fitted central slope k also increased, indicating that stronger competition produces a narrower transition region and therefore weaker spatial coexistence (Figure 5).

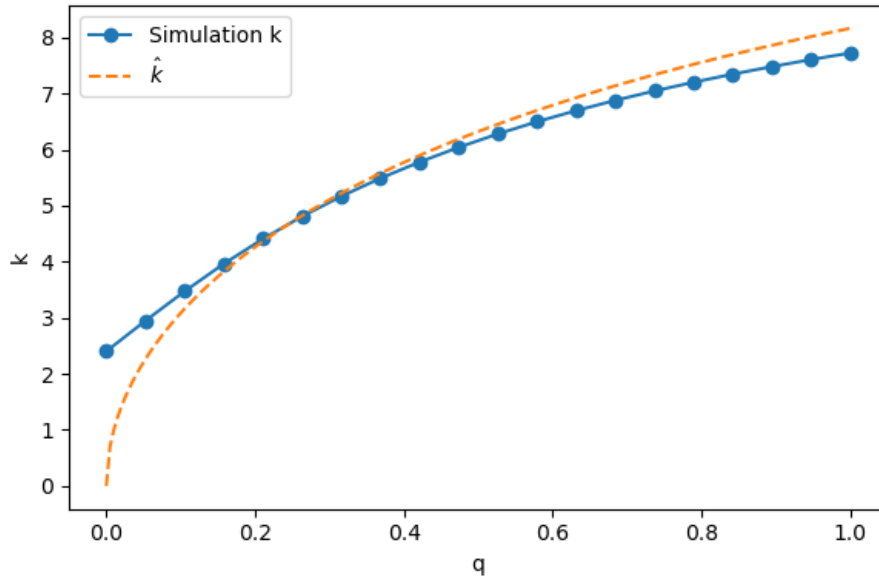


Figure 5: Relationship between the simulated central slope k and the theoretical estimate $\hat{k}$ as a function of $q = a - 1$. The approximation agrees well with simulations for weak to moderate competition strengths, but systematically underestimates k when q becomes large.

However, the Allen–Cahn approximation provides a good description only when q is around in $(0.2, 0.6)$. Substitute q into equation 24,

$$r = \frac{a - 1}{a + 1} = \frac{q}{q + 2}.$$

For small q , the effective coefficient approaches zero, yielding

$$\hat{k}|_{q \rightarrow 0} = \sqrt{2r} \rightarrow 0,$$

which is consistent with the broad transition fronts observed numerically. However, k is still

kept in simulation, with a value between 2 to 3, which might because of the initial state still kept. Besides, when $a \rightarrow \infty$, that is $q \rightarrow \infty$ the interface width should vanish and thus $k \rightarrow \infty$, whereas the approximation predicts only

$$\hat{k}|_{q \rightarrow \infty} \rightarrow \sqrt{2}.$$

This discrepancy indicates that the Allen–Cahn reduction is accurate only in the weak-competition regime and breaks down for strong interspecific interactions.

3.2 Balance Between Dispersal, Reproduction, and Competition

In more general parameter regimes, the two-species system does not necessarily converge to a stationary transition front. We first investigated the combined effects of diffusion and intrinsic growth by varying the diffusion coefficient D and intrinsic growth rate r over the interval $(0, 2)$. Simulations were performed for 100000 time steps with temporal step size 1×10^{-6} . The direction and speed of the transition front were characterized by the position x^* satisfying

$$N_1(x^*) = N_2(x^*).$$

A balanced coexistence state corresponds to the condition $x^* = 0$, indicating that the interface remains stationary throughout the simulation.

The resulting phase diagram reveals a positive relationship between diffusion rate and intrinsic growth rate along the balance condition. Species with higher dispersal ability therefore require correspondingly larger growth rates in order to maintain a stable spatial coexistence front (Figure 6A). Besides the balance condition, larger diffusion rate of species 2 lead to the dominance of species 1, while a larger intrinsic growth rate makes the dominance of species 2. Also, higher the two parameters is, the faster that the coexistence broken. This behavior differs fundamentally from the classical locally mixed coexistence condition in the generalized Lotka–Volterra model, where coexistence is determined solely by interaction coefficients.

To further characterize the balance relation between dispersal and reproduction, we analyzed the phase boundary in logarithmic coordinates. The balance condition appears approximately linear in the $\log D$ – $\log r$ plane, with slope close to 2 (Figure 6B). This observation suggests the scaling relation

$$D \sim r^2.$$

Therefore, the maintenance of a stationary coexistence front requires diffusion and local population growth to satisfy an approximate quadratic balance relation.

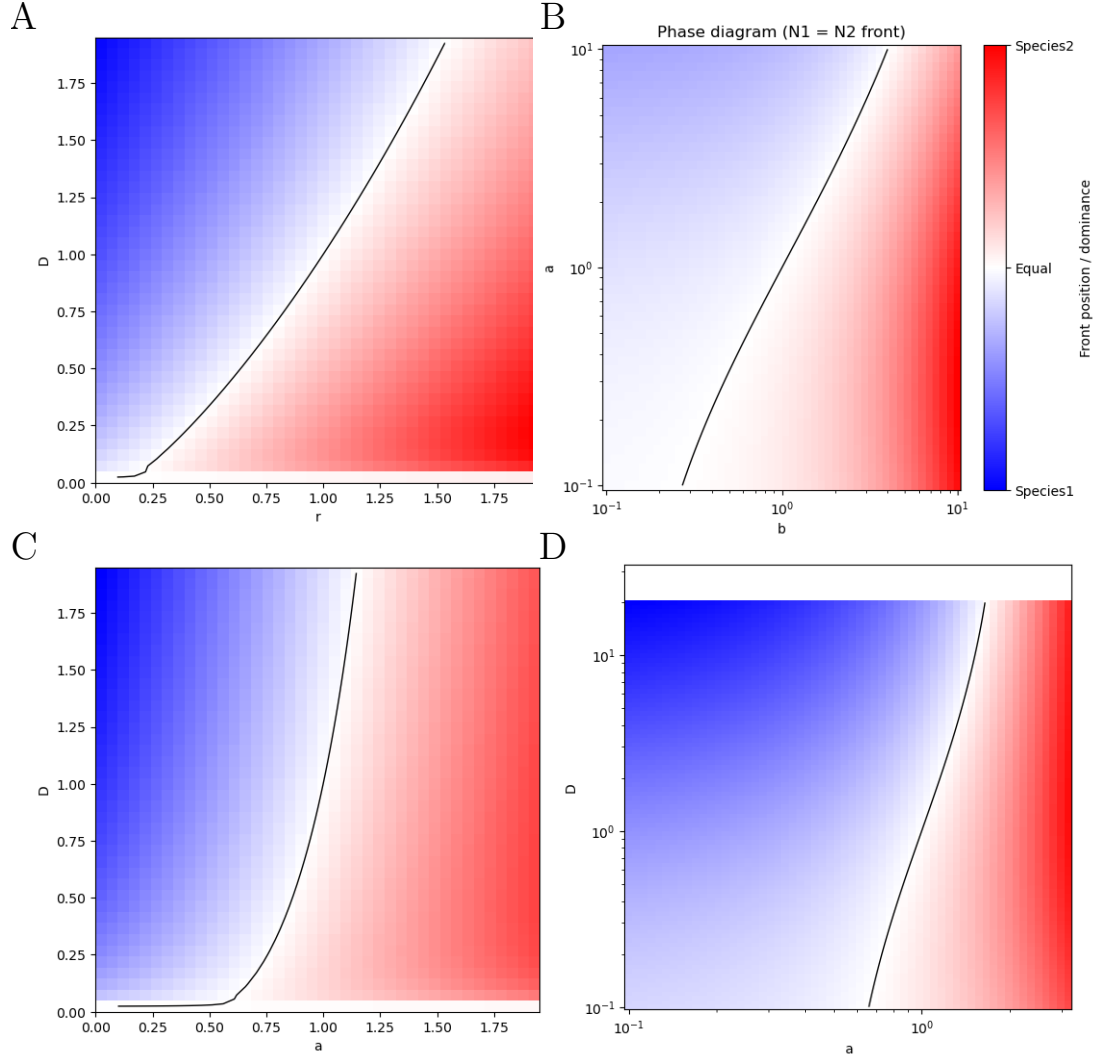


Figure 6: Transition dynamics and balance conditions under different parameter combinations. (A) Phase diagram in the (r, D) plane on linear scales. Blue regions indicate parameter combinations for which species 1 expands, whereas red regions indicate expansion of species 2. Lighter colors correspond to slower front velocities. The black curve denotes the balance condition at which the transition front remains stationary. (B) The same (r, D) phase diagram represented on logarithmic scales. (C) Phase diagram in the (a, D) plane on linear scales. (D) The corresponding (a, D) phase diagram on logarithmic scales.

For parameter combinations close to the balance boundary, the two species appear to coexist for long periods of time, although the interface position satisfying $N_1 = N_2$ is displaced from the center of the domain. However, long-term simulations reveal that this apparent coexistence is only transient. To illustrate this behavior, we simulated the case $D = 1.25$ and $r = 0.75$, corresponding to a parameter combination located slightly above the balance line (in figure 6A). In this regime, species 1 eventually takes over the entire domain and species 2 becomes extinct, though taking more than 1500000 steps (Figure 7A).

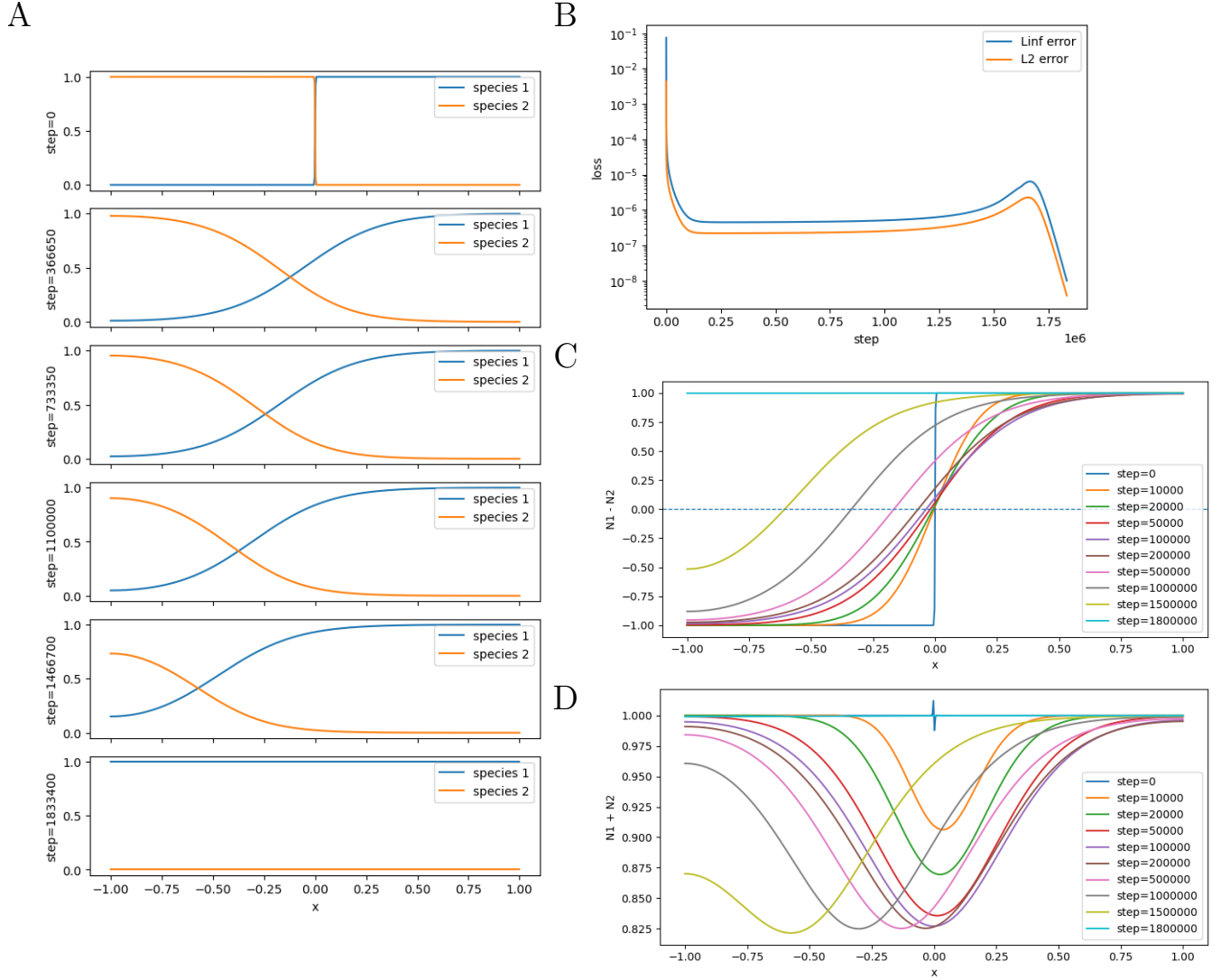


Figure 7: Simulation of an asymmetric two-species system with $D = 1.25$ and $r = 0.75$. (A) Spatial population dynamics. A transition front first forms and subsequently propagates toward the left boundary, eventually leading to the extinction of species 2. (B) Convergence error during the simulation. The error initially decreases rapidly, then remains near 10^{-6} during front propagation, exhibits a transient peak near the boundary, and finally decays after competitive exclusion is completed. (C) Temporal evolution of $N_1 - N_2$. A logistic-like transition front forms initially and subsequently translates leftward. (D) Temporal evolution of the total population density $N_1 + N_2$.

The convergence dynamics also differ substantially from those observed in the symmetric case. Initially, the simulation error decreases rapidly as the sharp initial boundary relaxes into a smooth transition front, similar to the balanced symmetric system. After the front is established, however, the error remains approximately constant over an extended period (Figure 7B). This plateau corresponds to the slow propagation of the interface across the domain: although the

front translates spatially, its shape remains nearly unchanged, resulting in only small differences between consecutive simulation steps. As the moving front approaches the domain boundary, the population distribution changes rapidly because of the no-flux boundary condition, leading to a temporary increase in the simulation error. Finally, once species 2 is eliminated, the system converges toward the single-species equilibrium and the error decreases rapidly again. These dynamics are analogous to those of an asymmetric Allen–Cahn system, in which the two stable states possess unequal potentials and the interface therefore propagates toward the energetically unfavorable state.

A similar analysis was performed for the balance relation between diffusion rate D and competition strength a . In a locally mixed competitive community, species with inter-specific competition coefficient smaller than unity are generally excluded by stronger competitors. In contrast, the spatial model reveals that coexistence can still emerge if the weaker competitor disperses sufficiently slowly. Specifically, species with lower competitive ability were able to coexist with a species whose competition coefficient was fixed at 1.5, provided that their diffusion rate was sufficiently reduced (Figure 6C). Moreover, the balance relation between D and a is also positive: stronger competitive interactions require larger dispersal rates to maintain a stationary coexistence front. In logarithmic coordinates, the phase boundary approximately follows a power-law relation with exponent between 3 and 4 (Figure 6D).

3.3 A Special Case of Multi-species Competitive Communities

For multi-species systems, the dynamics become substantially more complex and require a much larger parameter space to be explored numerically. To obtain a qualitative understanding of transition fronts in multi-species communities, we therefore focus on several simplified cases. Although these assumptions are not fully realistic ecologically, they allow the existence and structure of stable transition fronts to be analyzed more clearly.

First, we consider communities with purely competitive interactions. Restricting interactions to competition suppresses oscillatory or chaotic dynamics near equilibrium and constrains the total biomass of the system. Second, we assume that all species are identical in their dispersal and intrinsic growth rates, namely

$$D_i = 1, \quad r_i = 1.$$

Under these assumptions, we introduce a minimal toy model of multi-species bistability.

3.3.1 Approximate solution

To generate bistable community dynamics at the global scale, we partition the species into two competing sub-communities. Species within the same sub-community interact with strength $a = 1$, whereas species belonging to different sub-communities compete more strongly, with interaction strength $b > 1$. The resulting interaction matrix takes the form

$$\begin{pmatrix} 1 & \dots & a & b & \dots & b \\ \dots & \dots & \dots & \dots & \dots & \dots \\ a & \dots & 1 & b & \dots & b \\ b & \dots & b & 1 & \dots & a \\ \dots & \dots & \dots & \dots & \dots & \dots \\ b & \dots & b & a & \dots & 1 \end{pmatrix}.$$

Within each local sub-community, the generalized Lotka–Volterra dynamics reduce to

$$\frac{dN_i}{dt} = N_i \left(1 - \sum_{j=1}^S N_j \right). \quad (26)$$

At equilibrium, the total biomass satisfies

$$\sum_{j=1}^S N_j^* = 1,$$

whereas the equilibrium abundance of each individual species depends on the initial condition. Biologically, this degeneracy arises because species within the same sub-community are dynamically equivalent and therefore behave effectively as different components of the same functional group.

Let $N_{1,i}$ denote species belonging to sub-community 1, and $N_{2,j}$ species belonging to sub-community 2. The corresponding diffusion–reaction system is

$$\begin{aligned} \frac{\partial N_{1,i}}{\partial t} &= \nabla^2 N_{1,i} + N_{1,i} \left(1 - \sum_k N_{1,k} - \sum_\ell b N_{2,\ell} \right), \\ \frac{\partial N_{2,j}}{\partial t} &= \nabla^2 N_{2,j} + N_{2,j} \left(1 - \sum_\ell N_{2,\ell} - \sum_k b N_{1,k} \right). \end{aligned}$$

Because all species within the same sub-community experience identical interaction terms, the system can be reduced to the dynamics of the total sub-community biomasses,

$$C_n = \sum_i N_{n,i}.$$

The governing equations become

$$\begin{aligned}\frac{\partial C_1}{\partial t} &= \nabla^2 C_1 + C_1 (1 - C_1 - bC_2), \\ \frac{\partial C_2}{\partial t} &= \nabla^2 C_2 + C_2 (1 - C_2 - bC_1).\end{aligned}$$

This reduced system is mathematically identical to the symmetric two-species model analyzed previously, with the total sub-community biomasses replacing the species abundances. Consequently, the difference between the two sub-communities is also expected to exhibit an approximately logistic transition profile,

$$C_1(x) - C_2(x) \approx \frac{1}{1 + e^{\sqrt{2\frac{b-1}{b+1}}x}}. \quad (27)$$

Since the abundance of each species remains proportional to the total biomass of its sub-community, the spatial distributions of individual species are likewise expected to follow approximate logistic forms.

3.3.2 Numerical solution

We applied the IMEX semi-implicit method to simulate a six-species system. The first three species belong to stable community 1, whereas the remaining three species belong to stable community 2. The inter-community interaction strength was set to $b = 1.5$, while the intra-community interaction strength was fixed at $a = 1$. The two communities were assumed to be symmetric, with initial relative population sizes of the three species within each community given by 0.5, 0.3, and 0.2, respectively.

A clear and stable transition interface between the two communities was maintained once the population distributions converged (i.e., when the simulation loss decreased below the pre-defined threshold). All six species coexisted within the transition region (Figure 8A). In addition, within each community, the equilibrium abundance of each species remained proportional to its initial abundance.

The contrast between the two communities, measured by $C_1 - C_2$, converged to an approximately logistic distribution (Figure 8B), exhibiting a pattern similar to that observed in the symmetric two-species system. Likewise, the total biomass, $C_1 + C_2$, showed dynamics comparable to the two-species case: the total abundance remained close to 1 throughout the spatial domain, but exhibited a stronger deviation near the center of the transition region (Figure 8C).

3.3 A Special Case of Multi-species Competitive Communities

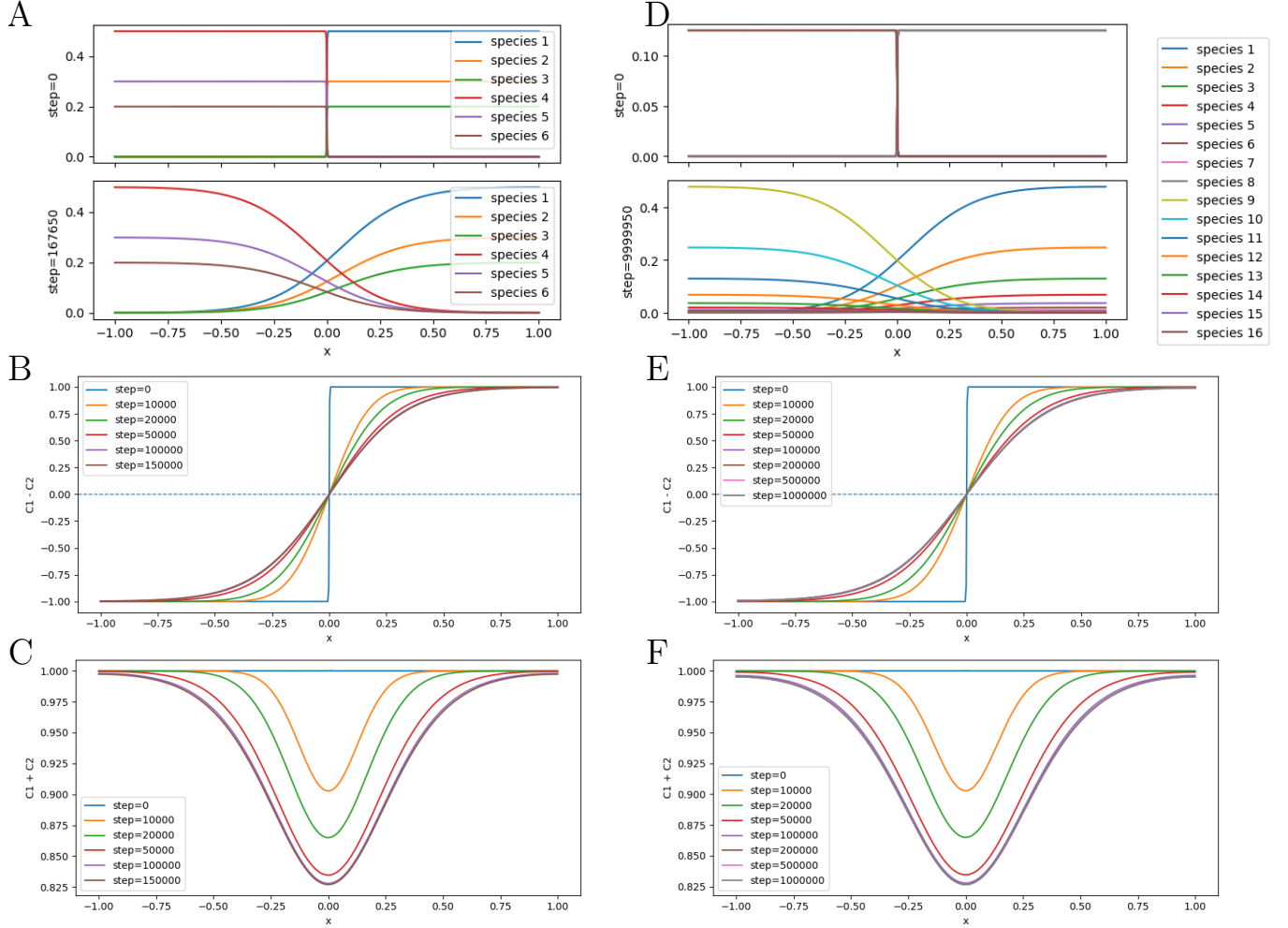


Figure 8: Simulations of multispecies competitive dynamics. (A) Spatial dynamics of a six-species community with intra-community interaction strength $a = 1$, inter-community interaction strength $b = 1.5$, and identical dispersal and growth rates for all species. A stable transition front forms, and the final species abundances depend on the initial population distribution. (B,C) Spatial distributions of $C_1 - C_2$ and $C_1 + C_2$ during the simulation, showing dynamics analogous to those of the symmetric two-species system. (D) Simulation of a sixteen-species community. The two subcommunities remain interactionally symmetric, with the same values of a and b as in panel (A), but species are assigned different dispersal rates: $D = 1, 1.05, 1.1, 1.15, 1.2, 1.25, 1.3, 1.35$ for species 1–8 and 9–16, respectively. The final spatial distribution depends strongly on dispersal ability. (E,F) Spatial distributions of $C_1 - C_2$ and $C_1 + C_2$ corresponding to the simulation shown in panel (D).

3.4 Logistic Equilibrium under Strong Inter-Community Competition

Building upon the simplified toy model described above, we next investigated the effects of heterogeneous dispersal rates. This simulation was designed to examine the robustness of the logistic-like transition structure under moderate variation in dispersal ability.

We considered a system in which the two local sub-communities remained interactionally symmetric ($a = 1$, $b = 1.5$). Each community contained eight species, with relative diffusion coefficients assigned as $D_i = 1, 1.05, 1.1, 1.15, 1.2, 1.25, 1.3, 1.35$.

All species persisted throughout the simulation, at least over the long simulation timescale considered here. The equilibrium abundance of each species within a community was negatively associated with its dispersal ability: species with lower diffusion rates maintained higher equilibrium population sizes (Figure 8D).

In addition, the community contrast, measured by $C_1 - C_2$ (Figure 8E), as well as the total biomass, $C_1 + C_2$ (Figure 8F), exhibited spatial patterns similar to those observed in the symmetric identical-species system. These results suggest that moderate heterogeneity in dispersal rates does not qualitatively alter the existence or overall structure of the stable transition interface.

3.5 Measure of β -Diversity under Logistic Spatial Distributions

The simulations described above provide a natural framework for measuring biodiversity using continuous spatial distributions rather than discrete community samples. In particular, the continuous β -diversity measure can be directly applied to spatial population distributions, such as the multispecies system with heterogeneous dispersal rates shown in Figure 8D.

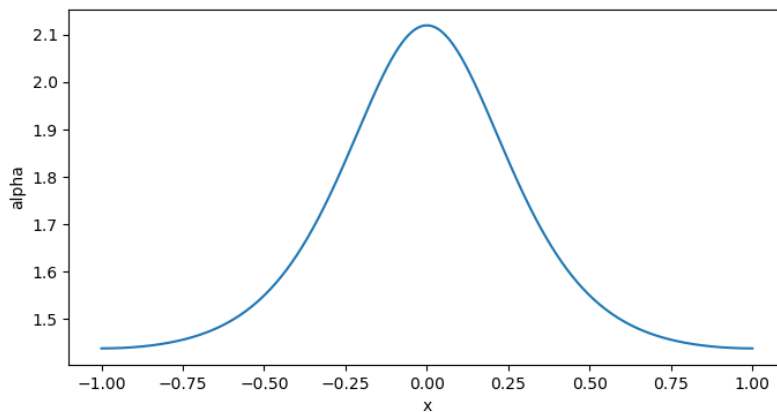


Figure 9: Spatial distribution of α -diversity across the transition region. It reached a maximum in the middle ($x = 0$), showing a symmetric pattern.

First, the spatial distribution of α -diversity was calculated across the simulated region (Fig-

ure 9). The α -diversity reached its maximum near the center of the transition zone, where species from the two alternative communities overlapped most strongly. By integrating $\alpha(x)$ over the spatial domain, the spatially averaged local diversity was obtained as

$$\bar{\alpha} = 1.654.$$

The average abundance of each species, denoted by $\bar{N}_i$, was then calculated by integrating the population distributions over space. These values were subsequently used to compute the global diversity,

$$\gamma = 2.115.$$

Consequently, the resulting β -diversity was

$$\beta = \frac{\gamma}{\bar{\alpha}} = 1.279,$$

which represents the theoretical β -diversity calculated directly from the continuous equilibrium distributions.

3.5.1 Comparison of β -diversity measures

To evaluate the performance of the proposed continuous β -diversity measure, we generated discrete observational samples from the simulated spatial distributions. An example of a sampled dataset is shown in Figure 10, where $k = 10$ sampling locations and $n = 100$ individuals per location were used.

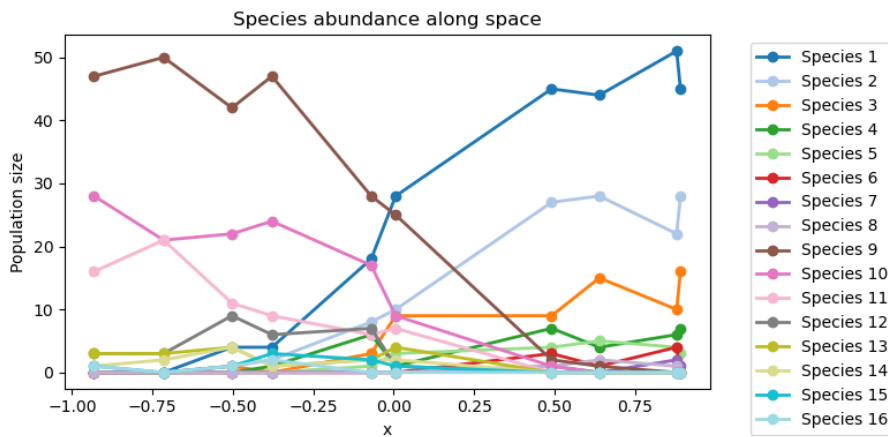


Figure 10: An example sample generated from the multi-species system with heterogeneous dispersal rates, using $k = 10$ sampling locations and $n = 100$ individuals per location.

Both the discrete and continuous β -diversity measures were calculated from these sampled data. Simulations were conducted for sampling strategies with $k = 5, 10$ and $n = 20, 50, 100$.

For each parameter combination, 50 independent samples were generated (Figure 11).

The mean values obtained from the two methods differed significantly in all sampling strategies ($p < 0.05$). The continuous measure generally produced estimates closer to the theoretical β value, particularly for the low-sampling condition ($k = 5$, $n = 20$) and for all cases with $k = 10$. However, the variances of the two methods were not significantly different in most cases (except for $k = 10$, $n = 50$), indicating that the continuous approach did not consistently reduce estimation variability.

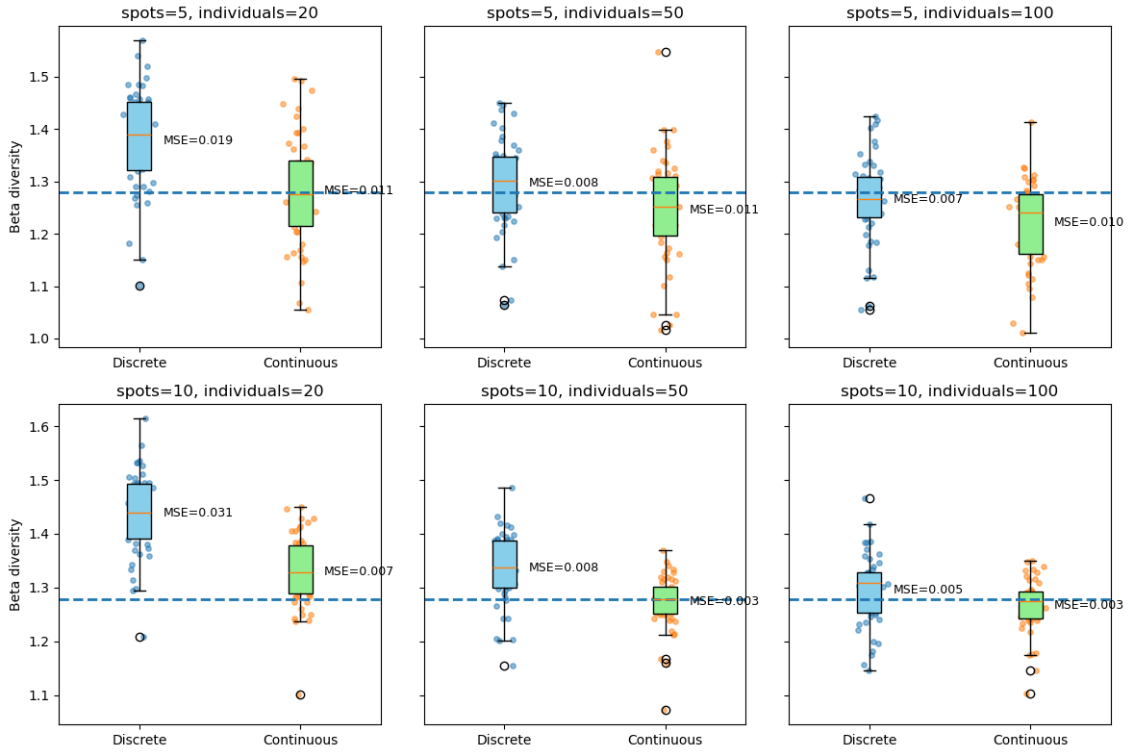


Figure 11: Comparison between discrete and continuous β -diversity measures under different sampling strategies. The number of sampling locations was set to $k = 5$ or 10, and the sample size per location was $n = 20, 50$, or 100. Each strategy was replicated 50 times. The dashed blue line shows the theoretical value from the simulation. The continuous method generally produced more accurate estimates, particularly when the number of sampling locations was large. Both methods improved with increasing sampling effort.

The mean squared error (MSE) of both methods decreased with increasing sampling effort, including both the number of sampling locations and the number of individuals sampled per location. Notably, the continuous measure consistently achieved lower MSE values when $k = 10$, suggesting that incorporating the fitted spatial distribution improves estimation accuracy when sufficient spatial information is available.

4 Discussion

4.1 Balance Conditions and Ecological Trade-offs

In island biogeography (MacArthur and Wilson, 2001) and meta-community theory (Hubbell, 2001, Loreau et al., 2003), trade-offs between dispersal ability and colonization or competitive performance are commonly considered a key mechanism promoting species coexistence. Species with high dispersal ability are generally assumed to be weaker local competitors, allowing inferior competitors to persist through repeated colonization of empty or disturbed patches. The central idea is that species with poor local competitive ability may still survive regionally if they can disperse efficiently among habitats.

However, the continuous spatial model studied here suggests a somewhat different balance condition. Simulations of interface movement under varying diffusion coefficients D and intrinsic growth rates r indicate that species with higher dispersal rates must also possess higher intrinsic growth rates in order to maintain a stable coexistence interface. Similarly, the D – a phase diagram shows that faster-dispersing species must exhibit weaker competitive ability to preserve the balance between the two communities. Comparable phenomena have also been reported in invasion models involving competing species expanding into open space (Girardin and Lam, 2019).

Mechanistically, during the invasion process, faster-dispersing species initially colonize the opposite spatial domain more rapidly (Namba and Mimura, 1980). However, if their intrinsic growth rate is insufficient, they may fail to establish stable populations after dispersal and eventually decline, because dispersal simultaneously reduces their abundance within their original domain (Dockery et al., 1998). Consequently, a dispersal advantage alone is not sufficient for long-term persistence; rapid growth is also required to maintain population density after colonization. In contrast, weaker competitive ability can counterbalance this advantage and prevent rapid competitive exclusion of the slower species.

These results suggest that the ecological concept of “colonization ability” may require reinterpretation in continuous spatial systems. Rather than representing dispersal alone, colonization ability may emerge from a combination of dispersal capacity, intrinsic growth, and competitive interactions. In particular, intrinsic growth rate should be viewed as an essential component determining whether a species can not only reach new habitats, but also successfully establish and maintain populations there.

Interestingly, in the present continuous model, lower dispersal rates are often intrinsically advantageous (Risler, 2017). In the D – r and D – a phase diagrams, the region in the left of the balance condition is dominated by species with lower diffusion rates (Figure 6A,C). This result is consistent with recent theoretical studies showing that, under strong competition, slower

dispersers may outcompete faster dispersers even when the species are otherwise symmetric (Nakamura and Ogiwara, 2025).

4.2 Generalization of Logistic Transition Fronts

Throughout our analysis, it is evident that logistic functions provide an excellent approximation for the shapes of species-composition transition fronts in the simplified models considered here. For symmetric competitive systems, this result is relatively intuitive, since the dynamics possess two symmetric stable equilibria (Figure 3C). Returning to the Allen–Cahn framework (Allen and Cahn, 1975), it is therefore reasonable to conjecture that reaction–diffusion systems with simple bistable dynamics and symmetric stable states naturally generate stationary traveling-wave solutions with zero propagation velocity.

In contrast, asymmetric systems generally lead to asymmetric stable equilibria, which often prevents the existence of stationary interfaces (Gardner, 1982, Hou and Leung, 2008). Nevertheless, our simulations reveal that certain parameter combinations can still support stable transition fronts even in asymmetric systems. This phenomenon was unexpected and may potentially be explained from the perspective of effective potential functions or energetic balance between the competing stable states. Further theoretical analysis is required to clarify the underlying mechanism.

More generally, however, the existence of stable interfaces appears to be highly sensitive to parameter values (Kan-on, 1995). In the parameter phase space, stationary fronts are often associated with unstable balance conditions, such that even small perturbations of system parameters may destabilize the interface and lead to directional propagation of the front. Despite this sensitivity, nearly symmetric systems can still maintain long-lived transition regions over ecologically relevant timescales, even if the interface eventually collapses in the long run.

Although stable fronts occupy only narrow regions of parameter space in the deterministic models considered here, realistic ecological systems are inevitably influenced by stochastic effects and environmental heterogeneity. Such fluctuations may effectively broaden the range of parameters under which transition fronts can persist, thereby increasing the likelihood of observing stable or quasi-stable coexistence interfaces in natural communities.

4.3 Equilibria of Bistable Communities with More General Interactions

Another important limitation of the present framework is that the models considered here are based exclusively on competitive interactions. The primary advantage of this assumption is that it allows the construction of highly simplified bistable multispecies systems: bistability can be generated simply by distinguishing between intra-community and inter-community competition

strengths. However, more general ecological systems are likely to exhibit substantially richer interaction structures.

On the one hand, ecological interactions are not restricted to competition alone. In this sense, the models analyzed here may be interpreted more specifically as describing dynamics within a single trophic level or functional group, such as competition among tree species in forest communities. Previous studies have already extended simple two-species competition systems by introducing additional trophic interactions, for example through the inclusion of predators in three-species models (Xie, 2012, Lv et al., 2013, Parshad et al., 2014). A more general and ecologically realistic extension would be to consider bistable random communities with heterogeneous interaction networks.

One possible approach for constructing such systems is to assemble two locally stable communities and impose sufficiently strong interactions between them. Based on the results obtained here, we conjecture that these systems may also generate logistic-like transition fronts, even when the detailed interaction structure is substantially more complex than the simplified competitive models considered in this work.

On the other hand, the bistability in our model was imposed in a highly explicit and idealized manner. In realistic random ecological communities, bistability may instead emerge spontaneously from the collective interaction structure without requiring artificially separated community modules (Maynard et al., 2018, Song and Saavedra, 2018, Angulo et al., 2021). In such cases, an important open question concerns the shape and stability of the resulting spatial interface.

From this perspective, it is useful to reconsider why logistic-like profiles emerge in the present models. Logistic functions capture two essential characteristics of the transition front: a sharp but continuous change near the center of the interface, together with monotonic exponential convergence toward stable equilibria on both sides. This observation suggests that logistic-like fronts may arise only when trajectories approach the stable equilibria asymptotically without oscillatory behavior in phase space (Rosen, 1977, Hou and Leung, 2008).

Conversely, if the local dynamics near equilibrium contain rotational or oscillatory components, more complicated interface structures may emerge. In particular, the edges of the transition zone may exhibit non-monotonic spatial patterns, oscillatory tails, or even spatially heterogeneous structures that cannot be approximated by simple logistic functions.

4.4 Limited Dispersal

In the present study, species dispersal was modeled using the classical diffusion operator, represented by the second spatial derivative. This formulation implicitly assumes random diffusion, in which the dispersal flux is proportional to the local population gradient. Although this

approximation is mathematically convenient and widely used in reaction–diffusion ecology, it also introduces an important physical and ecological limitation.

Specifically, classical diffusion equations formally permit infinitely fast propagation. In other words, a small fraction of individuals or particles may instantaneously disperse arbitrarily far over an arbitrarily short time interval. Such behavior is unrealistic for biological organisms, whose movement is always constrained by finite dispersal velocities and environmental limitations.

One possible approach for resolving this issue is to replace linear diffusion with nonlinear density-dependent dispersal models. For example, the diffusion coefficient may itself depend on local population density (Okubo and Chiang, 1974, Namba, 1980),

$$\partial_t u = \nabla (D(u) \nabla u), \quad D(u) = \frac{u^b}{(1-u)^b}.$$

In such models, dispersal becomes strongly dependent on local abundance. In particular, the diffusion coefficient approaches zero as population density approaches zero, preventing isolated individuals from dispersing arbitrarily rapidly into empty regions (Figure 12A). Consequently, the propagation speed of the front becomes effectively bounded, leading to more realistic finite-velocity dispersal dynamics (Wang et al., 2024) (Figure 12A).

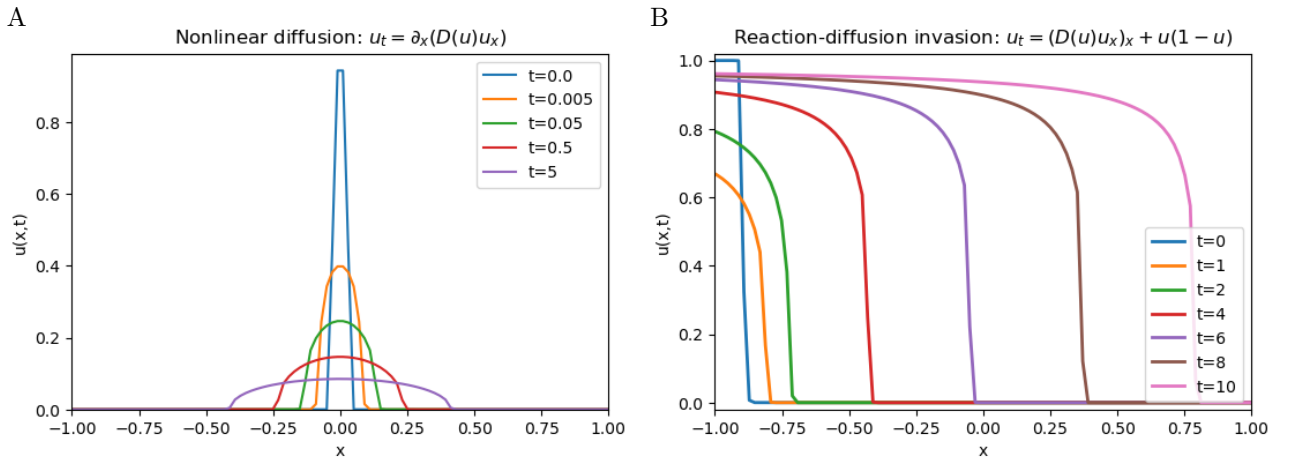


Figure 12: A. The nonlinear diffusion from a central narrow initial distribution. Note the solid end. B. Reaction-diffusion model with nonlinear diffusion, there is still a constant speed.

More generally, nonlinear diffusion may substantially alter the geometry and stability of ecological transition fronts. In contrast to classical Fisher–KPP or Allen–Cahn systems, density-dependent diffusion can generate sharper interfaces, compactly supported traveling waves, or front pinning phenomena.

Another important issue concerns the assumption that dispersal is purely random. In many

ecological systems, both the magnitude and direction of dispersal may depend on local ecological conditions rather than following unbiased diffusion alone.

On the one hand, species interactions themselves may influence dispersal behavior. Individuals may preferentially move toward regions with higher growth potential or lower competitive pressure, resulting in directed dispersal rather than purely random movement. Such processes can be incorporated into the present framework using density-dependent or interaction-dependent diffusion terms (Rodrigo and Mimura, 2001, Wang and Guo, 2023), similar to the nonlinear diffusion models discussed in the previous section. In general, these mechanisms are expected to generate sharper and more localized transition fronts because dispersal becomes coupled to local ecological fitness.

On the other hand, environmental gradients, including variations in temperature, moisture, nutrient availability, or other abiotic resources, can strongly influence species dispersal, growth, and interaction dynamics (Hader et al., 1974, Amarasekare, 2003). Spatial environmental heterogeneity alone may therefore generate changes in community composition, which forms the basis of most empirical studies of β -diversity and species turnover.

The present work should be viewed as complementary to this traditional perspective. Our results demonstrate that even in the absence of environmental gradients, the combined effects of dispersal and species interactions are sufficient to generate stable or long-lived compositional transition fronts. Consequently, purely endogenous ecological dynamics may contribute substantially to spatial species turnover and therefore to observed β -diversity patterns.

This observation suggests that measurements of β -diversity should account not only for environmentally driven turnover, but also for transition structures generated intrinsically by ecological interactions and dispersal processes. In particular, it may be important to distinguish environmentally induced compositional changes from those arising through approximately neutral or self-organized spatial dynamics.

4.5 Pattern Formation in Ecology

Reaction–diffusion models were originally developed to study biological pattern formation during development (Turing, 1990) and were later generalized to a broad range of morphogenetic processes (Kondo and Miura, 2010). Classical applications include mammalian coat patterns (Sick et al., 2006, Kaelin et al., 2012, Yamaguchi et al., 2007), floral pigmentation patterns in plants (Liang et al., 2023), and even the morphogenesis of organs such as the liver (De Langhe et al., 2005), kidney (Menshykau and Iber, 2013), and limbs (Raspopovic et al., 2014, Onimaru et al., 2015). Despite their mathematical simplicity, these models are capable of generating remarkably complex spatial structures through the interaction between local reactions and dispersal processes.

More recently, reaction–diffusion frameworks have also been increasingly applied in ecology to explain the emergence of spatial organization in ecological communities (Parshad et al., 2014, Pichon et al., 2024). From this perspective, the present work may be interpreted not only as a model of stable boundaries between alternative communities, but also more generally as a mechanism for ecological patch formation at larger spatial scales.

To further explore this possibility, we first simulated a larger-scale system consisting of two symmetric competing species (Figure 13). The initial population distributions were generated from random normal fluctuations across space. The system developed alternating spatial domains dominated by different species, and these patterns could persist for a relatively long period, although the less abundant species eventually went extinct.

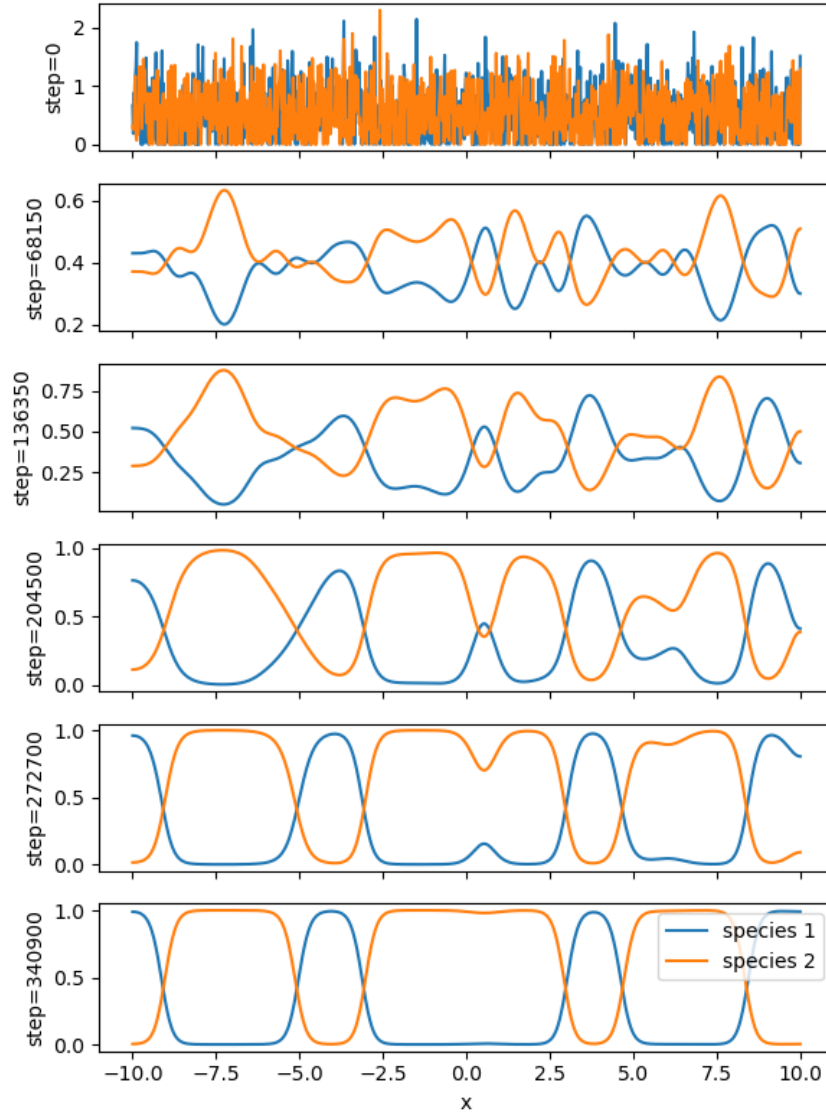


Figure 13: Interaction–dispersal dynamics in larger space with random initial state.

We further extended the model to two-dimensional space and simulated the dynamics from completely random initial community compositions (Figure 14). The resulting dynamics exhibited strong similarities to classical Turing-like pattern formation. Although the two competing species were assigned identical diffusion coefficients, causing the system to eventually converge toward relatively simple domains separated by stable interfaces, the transient dynamics nevertheless generated complex self-organized spatial structures. Moreover, when viewed across different temporal and spatial scales, or under the influence of stochastic fluctuations, the system is capable of producing a wide range of heterogeneous ecological patterns. These results suggest that even simple interaction–dispersal systems may contribute to large-scale spatial organization in ecological communities without requiring explicit environmental heterogeneity.

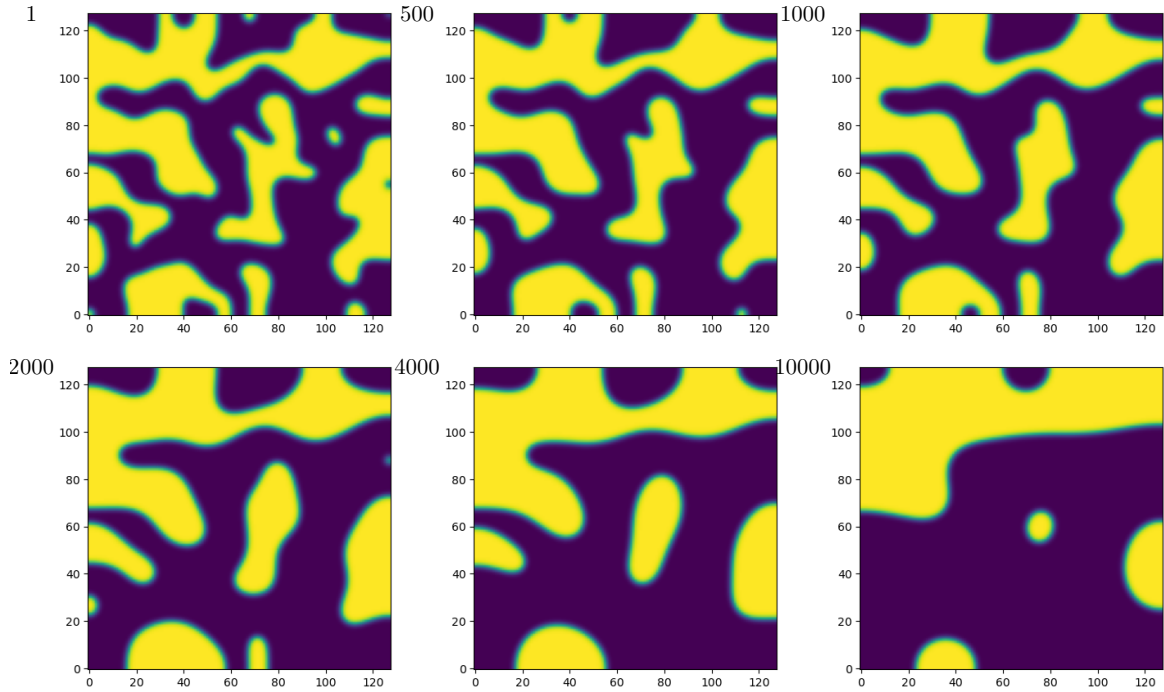


Figure 14: Interaction–dispersal dynamics in two-dimensional space. Colors indicate the regions dominated by each species. Simulations were initialized from spatially random community compositions.

4.6 Measuring β -diversity as Information Loss

β -diversity measures the variation or heterogeneity of species composition across space (Whittaker, 1960, Anderson et al., 2011), thereby linking local diversity patterns to biodiversity at larger spatial scales. Despite its central importance in community ecology, there is still no universally accepted theoretical framework for quantifying β -diversity. Numerous measures have been proposed, including Hill-number partitioning (Jost, 2007), β -deviation (Kraft et al., 2011), turnover–nestedness decomposition (Baselga, 2012), and variance-based approaches de-

rived from community composition matrices (Legendre and De Cáceres, 2013). However, many of these approaches implicitly underestimate the contribution of novel species combinations. In particular, a metacommunity in which no additional species are introduced, but where species assemble into new local combinations, may still exhibit substantial ecological heterogeneity despite having relatively low conventional β -diversity values.

Recently, Song et al. (2025) proposed a geometric interpretation of β -diversity. In this framework, each community composition is represented as a point in a high-dimensional space defined by species abundances, and β -diversity is quantified by the relative geometric volume occupied by observed communities. Although this framework was originally introduced to characterize the diversity of community states, it may also be interpreted from the perspective of information theory. Since both α - and γ -diversity can be understood as measures of information associated with species identity, β -diversity may naturally be interpreted as the information lost when ecological observations are aggregated from local to global scales. Under this interpretation, a novel species combination should contribute to β -diversity even if no new species are introduced, because such combinations increase local compositional information without changing the regional species pool.

In addition, most existing β -diversity measures are based on discrete observations and presence–absence data. However, many empirical studies investigate species turnover along continuous spatial gradients. Our framework therefore provides a simple null model for describing continuous species distributions, although currently under relatively restrictive assumptions. Based on this framework, discrete sampling data may be fitted to continuous spatial distributions, thereby reducing sampling bias and stochastic sampling effects. Our results suggest that fitting sampled data to continuous distributions can improve the estimation of species turnover (Figure 10), particularly when spatial sampling is sufficiently dense. In contrast, sparse sampling designs may not provide enough information to reliably estimate the underlying logistic structure.

From this perspective, the continuous β -diversity measure defined in Equations 18 and 19 can be rewritten as

$$\beta = L \frac{\sum_i q_i \ln(q_i)}{\int \sum_i p_i(x) \ln(p_i(x)) dx}, \quad p_i(x) = \frac{n_i(x)}{\sum_j n_j(x)}, \quad q_i = \frac{\int n_i(x) dx}{\sum_j \int n_j(x) dx}.$$

When the sampling size is constant across space, namely $\sum_i n_i(x) = n$ for all x , we obtain

$$p_i(x) = \frac{n_i(x)}{n}, \quad q_i = \frac{1}{L} \int p_i(x) dx,$$

and therefore

$$\beta = \frac{\sum_i \int p_i(x) dx \ln(q_i)}{\int (\sum_i p_i(x) \ln(p_i(x))) dx}. \quad (28)$$

This formulation resembles the structure of the Kullback–Leibler divergence (Kullback and Leibler, 1951), although expressed as a ratio rather than a difference. Conceptually, it measures the relative contribution of global entropy to local entropy across space.

Moreover, the spatial distribution of local entropy, represented here by α -diversity (Figure 9), reaches its maximum near the center of the transition region. This observation is consistent with the ecological expectation that biodiversity is elevated near the boundary between distinct communities, where species originating from both sub-communities coexist locally. In this sense, the transition front behaves as a “soft” ecological boundary rather than a sharp interface. Consequently, a large proportion of the observed β -diversity originates from the transition region itself. This interpretation further supports the importance of novel species combinations in characterizing spatial biodiversity patterns.

Finally, this framework may provide a basis for improved sampling strategies. Since regions with high local entropy contribute disproportionately to β -diversity, future approaches could incorporate adaptive or importance-sampling methods based on preliminary estimates of α -diversity in space.

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Code availability

Codes for generating the result in this thesis are available in Github ([coexist_in_space](#)).

A Generalized Lotka-Volterra Model

Population size N is the core quantity that we are interested when focusing on the dynamics of population and community. The state of community is then defined on that. $N = 0$ represent the extinction of certain species, and the stable state of community is the condition where $N = N^*$ do not change along time. To model the dynamics of population size N , one may apply a difference equation or differential equation. These model normally take two part of biological process into consideration, namely birth ($b(N)$) and death ($d(N)$). The difference equation model assume that both of the process of a population is discrete, for example, happened in certain season, while the differential equation model assume these biological process are continuous. Also, the later one could be viewed as a limit of the forth one when time interval approaching to 0.

A.1 One Species: Exponential, Logistic, and Varieties

In the simplest case, where both birth and death rate is a constant, the difference equation is written as,

$$N(t+1) - N(t) = bN(t) - dN(t),$$

which could be simplified as,

$$N(t+1) - N(t) = rN(t), \quad (29)$$

where $r = b - d$, the so called intrinsic rate. Then the solution of $N(t)$ would be

$$N(t) = N(0) \cdot (1 + r)^t$$

Also, for continuous model, it would be,

$$\frac{dN}{dt} = bN - dN = rN, \quad (30)$$

which has a solution after integral,

$$N(t) = N(0)e^{rt}.$$

Both of the models show that if $b > d$, the population size will get to infinity with an exponential increase. Also, notice here in both model we use b , d , and r , they have the same meaning but different measure in these two models.

These dynamics will get more interesting when we take environment capacity K into con-

sideration. Since the individuals of the same species are competing for same resource, there must be a capacity which is the maximal population size the habitat could stand. This competition can be modeled by the term $(1 - \frac{N}{K})$, indicate that it is increasing when population get close to the capacity. Now the difference model is,

$$N(t+1) - N(t) = rN(t)(1 - \frac{N(t)}{K}). \quad (31)$$

When this system get to an equilibrium N^* , that calls $N(t+1) = N(t)$, we have that $N^* = 0$ and $N^* = K$. $N^* = 0$ is instable and $N^* = K$ is stable. However, in discrete model, there might also be periotic or chaotic oscillations around the capacity because of the discrete step.

The continuous model,

$$\frac{dN}{dt} = rN(1 - \frac{N}{K}), \quad (32)$$

is known as the Logistic model. There are also two fixed point $N^* = 0, K$ that makes $\frac{dN}{dt} = 0$. Let $f(N) = rN(1 - \frac{N}{K})$, then $f'(n) = r - \frac{2r}{K}N$. Thus, at the fixed points, $f'(0) = r > 0$, $f'(K) = -r < 0$, which means that $N = 0$ is instable and $N = K$ is stable. This fact could also be shown in the phase graph.

Given a initial state $N(0) = N_0$, the dynamics could be fully solved.

$$\begin{aligned} \frac{dN}{dt} &= rN(1 - \frac{N}{K}) \\ \int \frac{K}{N(K-N)} dN &= \int r dt \\ \int \left(\frac{1}{N} - \frac{1}{N-K} \right) dN &= \int r dt \\ \ln \frac{N}{N-K} &= rt + C \\ \frac{N}{N-K} &= e^{rt+C} \end{aligned}$$

Then

$$\begin{aligned} N(t) &= \frac{K}{1 - e^{-rt-C}} \Rightarrow e^{-C} = 1 - \frac{K}{N_0} \\ N(0) &= \frac{K}{1 - e^{-C}} \\ N(t) &= \frac{K}{1 + (\frac{K}{N_0} - 1)e^{-rt}} \end{aligned}$$

Different to the discrete model, the dynamic is simple. Since $\frac{dN}{dt} > 0$ when $0 < N < K$, and $\frac{dN}{dt} < 0$ when $N > K$, the population size always converge to capacity K singularly.

There are amount of varieties of logistic model, by adding different effect to change the

intrinsic rate, for example, different Holling response functions. One of the cases that related to this project is Allee effect, which modeled the benefit of population density.

$$\frac{dN}{dt} = rN(1 - \frac{N}{K})(\frac{N}{A} - 1). \quad (33)$$

New parameter A is the critical point of population density. Survival rate is negative when $N < A$ and increased when $N > A$, because of some effect of high population density, for example, collaboration or mating probability. In this case, there are three fixed point $N^* = 0, k, A$. Similarly to the logistic model, we could find that $N^* = 0, K$ are stable and $N^* = A$ is instable. If initial population $N_0 > A$, the population size will converge to K as logistic model, with a lower speed; if $N_0 < A$, the population will die out.

A.2 Two Species: Lotka-Volterra, Competing, and Generality

When there are more than one species, the interaction between species must be considered. In the simplest case, two species, the system is written as,

$$\begin{aligned} \frac{dN_1}{dt} &= f(N_1, N_2) \\ \frac{dN_2}{dt} &= g(N_2, N_1) \end{aligned} \quad (34)$$

The Lotka-Volterra model, also known as the predator-prey model, describes the dynamic of such a system where one species is feed on the other one.

$$\begin{aligned} \frac{dP}{dt} &= \alpha P - \beta PA \\ \frac{dA}{dt} &= -\gamma A + \delta AP \end{aligned} \quad (35)$$

In this model, prey, or plant P , growth with a intrinsic birth rate α , and decline proportional to the meeting probability $P \cdot A$ with a rate β . The predator, or animal A , decreasing with a rate γ and survive by capture P with a rate δ . The dynamic could be illustrated in the P - A plane. On the fixed point, $\frac{dP}{dt} = 0$ and $\frac{dA}{dt} = 0$, the first calls $P = 0, \alpha = \beta A$, the later calls $A = 0, \gamma = \delta P$. Thus there are two fixed point $(P^*, A^*) = (0, 0), (\frac{\gamma}{\delta}, \frac{\alpha}{\beta})$. To analyse the stability of each point, we calculate the Jacobian matrix. Let $f = \alpha P - \beta PA$ and $g = -\gamma A + \delta AP$, then

$$\mathbf{J} = \begin{pmatrix} \frac{\partial f}{\partial P} & \frac{\partial f}{\partial A} \\ \frac{\partial g}{\partial P} & \frac{\partial g}{\partial A} \end{pmatrix},$$

which represent the dynamics around the fixed point. Around the fixed point, the dynamics of

small disturb $(\Delta P, \Delta A)$ could be linearized as

$$\frac{d}{dt} \begin{pmatrix} \Delta P \\ \Delta A \end{pmatrix} \approx \mathbf{J} \begin{pmatrix} \Delta P \\ \Delta A \end{pmatrix}$$

Let $\mathbf{x} = \begin{pmatrix} \Delta P \\ \Delta A \end{pmatrix}$, $\mathbf{x}(t) = e^{\mathbf{J}t} \mathbf{x}(0)$, which converge to 0 only when the eigenvalues of $\mathbf{J} < 0$. For this case,

$$\mathbf{J} = \begin{pmatrix} \alpha - \beta A & -\beta P \\ \delta A & -\gamma + \delta P \end{pmatrix},$$

at the fixed points,

$$\mathbf{J}|_{(0,0)} = \begin{pmatrix} \alpha & 0 \\ 0 & -\gamma \end{pmatrix}, \mathbf{J}|_{(\frac{\gamma}{\delta}, \frac{\alpha}{\beta})} = \begin{pmatrix} 0 & -\frac{\beta\gamma}{\delta} \\ \frac{\delta\alpha}{\beta} & 0 \end{pmatrix},$$

Thus, at $(0,0)$, $\lambda_1 = \alpha > 0$, $\lambda_2 = -\gamma < 0$ is instable. at $(\frac{\gamma}{\delta}, \frac{\alpha}{\beta})$, $\lambda_{1,2} = \pm\sqrt{\alpha\gamma}i$ are pure imagine numbers, this is a stable center, with orbit around it. As a result, in Lotka-Volterra model, both predator and prey are in periodic oscillation.

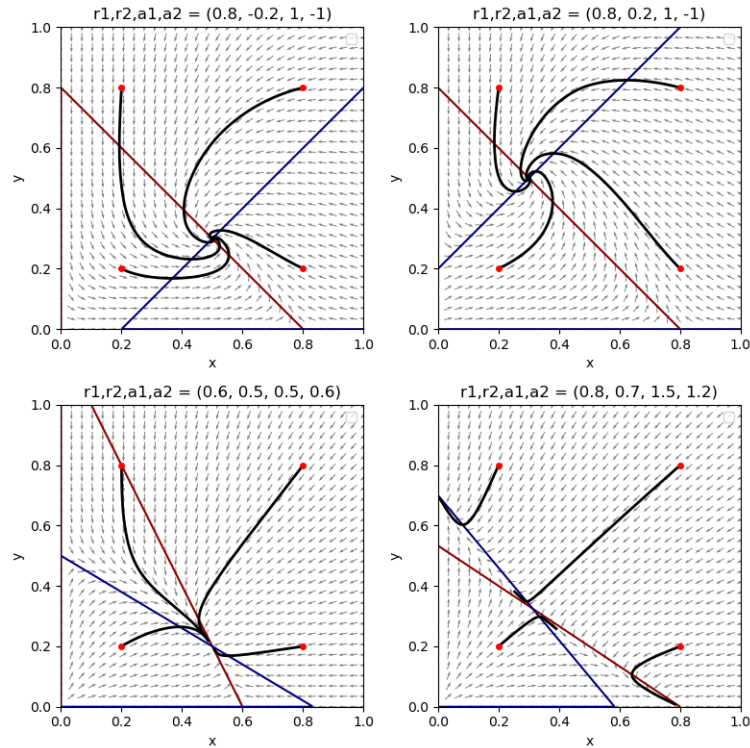


Figure S.1: For cases of two competing species. a_1 and a_2 represent $\alpha_{1,2}$ and $\alpha_{2,1}$. $\alpha_{i,i}$ is set to 1. The first three condition have a stable equilibrium, while the last one do not have a coexist equilibrium.

Another special case is two competing species N_1 and N_2 . Competing will decrease the growth rate proportional to the competing species,

$$\begin{aligned}\frac{dN_1}{dt} &= r_1 N_1 \left(1 - \frac{N_1 + \alpha_{1,2} N_2}{K_1}\right) \\ \frac{dN_2}{dt} &= r_2 N_2 \left(1 - \frac{N_2 + \alpha_{2,1} N_1}{K_2}\right),\end{aligned}\tag{36}$$

where $r_1 > 0$, $\alpha_{i,j}$ is the competing coefficient of species j to species i , compared to intra-specific competition. There are four fixed point, which are the nodes of nullclines $N_1 = 0$, $N_2 = 0$, $N_1 - \alpha_{1,2} N_2 = K_1$, $N_2 - \alpha_{2,1} N_1 = K_2$. The origin point $(0, 0)$ is always an instable node. The coexist node is $N_1^* = \frac{K_1 - \alpha_{1,2} K_2}{1 - \alpha_{1,2} \alpha_{2,1}}$, $N_2^* = \frac{K_2 - \alpha_{2,1} K_1}{1 - \alpha_{1,2} \alpha_{2,1}}$. There are also two boundary points $(K_1, 0)$, $(0, K_2)$. With different parameter combinations, there might be the following three conditions: two species coexist, only one species can take over, one species take over depending on the initial state.

More generally, a two species system can be modeled as,

$$\begin{aligned}\frac{dN_1}{dt} &= r_1 N_1 (1 + \alpha_{1,1} N_1 + \alpha_{1,2} N_2) \\ \frac{dN_2}{dt} &= r_2 N_2 (1 + \alpha_{2,2} N_2 + \alpha_{2,1} N_1),\end{aligned}\tag{37}$$

which include both inter- and intra- specific competition (Figure S.1).

A.3 Generalized Lotka-Volterra Model, and the Instability

Furthermore, following the same modelling approach above, if there are S different species in the community, S related differential equations are needed to model the dynamics. The generalized Lotka-Volterra model is:

$$\frac{dN_i}{dt} = r_i N_i \left(1 + \sum_j \alpha_{i,j} N_j\right),$$

or, more clear for the later analysis,

$$\frac{dN_i}{dt} = N_i \left(r_i + \sum_{j=1}^S A_{ij} N_j\right),\tag{38}$$

where $A_{ij} = r_i \cdot \alpha_{i,j}$. Population increase rate is now $\left(r_i + \sum_{j=1}^S \alpha_{i,j} N_j\right)$, include the intrinsic rate as well as effect of inter-specific interactions. Since the increase rate is a linear function of

population size, GLV could be write in the matrix form,

$$\frac{d\mathbf{N}}{dt} = \mathbf{D}(\mathbf{N}) (\mathbf{r} + \mathbf{A} \cdot \mathbf{N}), \quad (39)$$

where $\mathbf{A}$ si the interaction matrix. This matrix is normally be visualized as interaction network in empirical studies. The structure and pattern of the network might affect the dynamics of the system.

Such a system has 2^S equilibrium, which is the solution of either

$$\mathbf{D}(\mathbf{N}) = 0,$$

or

$$(\mathbf{r} + \mathbf{A} \cdot \mathbf{N}) = 0.$$

This system is called stable only when there is a positive equilibrium and is asymptotic stable. That is, all species exist and will not die out because of small disturb. Firstly, the existence of the equilibrium where all population size are positive is determined by the feasible condition,

$$\mathbf{N}^* = -\mathbf{A}^{-1}\mathbf{r} > 0, \quad (40)$$

which means that the inner equilibrium that no species die out is positive for every species.

Then, if such a equilibrium exist, the stability can be analysed by disturb analysis. If a small disturb $\mathbf{x}$ happened when system is at the equilibrium $\mathbf{N}^*$, the state would be $\mathbf{N} = \mathbf{N}^* + \mathbf{x}$. Let $\mathbf{f} = \mathbf{D}(\mathbf{N}) (\mathbf{r} + \mathbf{A} \cdot \mathbf{N})$,

$$\begin{aligned} \frac{d\mathbf{x}}{dt} &= \frac{d(\mathbf{N} - \mathbf{N}^*)}{dt} = \mathbf{D}(\mathbf{N}) (\mathbf{r} + \mathbf{A} \cdot \mathbf{N}) \\ &= \mathbf{D}(\mathbf{N}^* + \mathbf{x}) (\mathbf{r} + \mathbf{A} \cdot (\mathbf{N}^* + \mathbf{x})) \\ &= \mathbf{D}(\mathbf{N}^* + \mathbf{x}) \cdot \mathbf{A} \cdot \mathbf{x} \\ &= \mathbf{D}(\mathbf{N}^*) \cdot \mathbf{A} \cdot \mathbf{x} + \mathbf{D}(\mathbf{x}) \cdot \mathbf{A} \cdot \mathbf{x} \end{aligned}$$

The first order approximation is

$$\frac{d\mathbf{x}}{dt} \approx \mathbf{M} \cdot \mathbf{x} + \mathcal{O}(\mathbf{x}),$$

where $\mathbf{M} = \mathbf{D}(\mathbf{N}^*) \cdot \mathbf{A}$ thus,

$$\mathbf{x} \approx \mathbf{x}_0 \cdot e^{\mathbf{M}t}.$$

The small disturb around equilibrium $\mathbf{N}^*$ converge to 0 only if the real part of all the eigenvalues

of matrix M are smaller than 0, that is the stable condition,

$$\max_i \operatorname{Re} [\lambda_i(M)] < 0,$$

where $\lambda_i(M)$ is the eigenvalue of M . The same stable condition can also be deductive from Jacobian matrix $J(N)$, by defining $M = J(N^*)$.

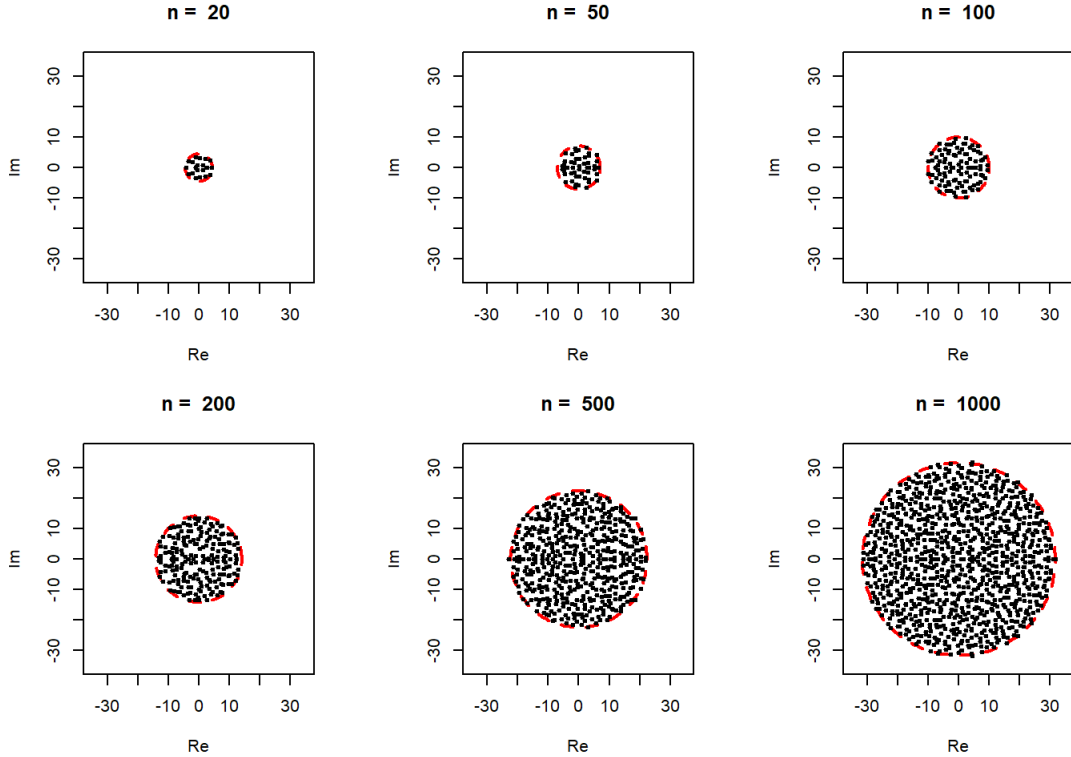


Figure S.2: The spectrum of normal distributed matrix.

Now we consider the stability of a random community. Ignoring the feasibility, we focus on the stability which related to the eigenvalues of Jacobian matrix. Now, if we assume that matrix M is totally random, where all the elements besides the diagonal follow a certain symmetric distribution, and the diagonal elements are $-k$. M can be divided as $M_1 - k \cdot I$. The eigenvalues of a random real-valued matrix are also solutions of a large linear function, as a result, they are complex numbers. The distribution of the eigenvalues of a matrix is the so called **Spectrum** of the matrix. And here is a circular law of random matrix: if the elements of a n by n random matrix follow the same symmetrical distribution with mean 0 and variance 1, with the increase of n , the distribution of eigenvalues λ_i divided by $\sqrt{n}$ converge to a uniform distribution in the unit circle. That is, the radius of the spectrum converge to $\sqrt{n}$ (Figure S.2).

$$\lim_{n \rightarrow \infty} r = \sqrt{n}. \quad (41)$$

The spectral radius is actually related to the mean strength (the value of elements). As a result, the radius will change if we change the mean strength, by decrease the variance, or decrease the elements density. If the variance of elements is $\mathbb{E}M_{ij}^2 = \sigma^2$, the radius will follow: $\lim_{n \rightarrow \infty} r = \sigma \cdot \sqrt{n}$. Beside, if we set the density of non-zero elements in a matrix is C (which means the connectedness of the network), the radius will follow: $\lim_{n \rightarrow \infty} r = \sqrt{C \cdot n}$. The general condition is

$$\lim_{n \rightarrow \infty} r = \sigma \cdot \sqrt{C \cdot n}. \quad (42)$$

As a result, matrix M is stable if the spectrum radius is smaller than k . However, the radius is increasing with the species number. Thus, a larger random community is less possible to be stable, as argued by May (1972).

B Fisher-KPP Equation

Here we do some basic analysis on the Fisher-KPP model (Fisher, 1937). The Fisher-KPP model is written as,

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + ru(1 - au)$$

Introduce the nondimensional variables

$$u' = au, \quad \tau = rt, \quad z = \sqrt{\frac{r}{D}} x.$$

Then the equation becomes

$$\frac{\partial u'}{\partial \tau} = \frac{\partial^2 u'}{\partial z^2} + u'(1 - u').$$

Therefore, it is sufficient to study the normalized Fisher equation

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + u(1 - u).$$

We look for travelling wave solutions whose profile remains unchanged during propagation. Let

$$z = x - ct,$$

where c is the wave speed. Substituting $u(x, t) = u(z)$ into the PDE yields

$$\frac{d^2 u}{dz^2} + c \frac{du}{dz} + u(1 - u) = 0.$$

The nonlinear term

$$f(u) = u(1 - u)$$

has two equilibria, $u = 0$ and $u = 1$ (Figure S.3A). The travelling wave solution connects these two states. Near $u = 0$, we linearize the equation by assuming

$$u(z) \sim e^{\lambda z}.$$

Substitution gives the characteristic equation

$$\lambda^2 + c\lambda + 1 = 0.$$

Hence,

$$\lambda = \frac{-c \pm \sqrt{c^2 - 4}}{2}.$$

To ensure decay as $z \rightarrow +\infty$, we require $\lambda < 0$, which implies

$$c \geq 2.$$

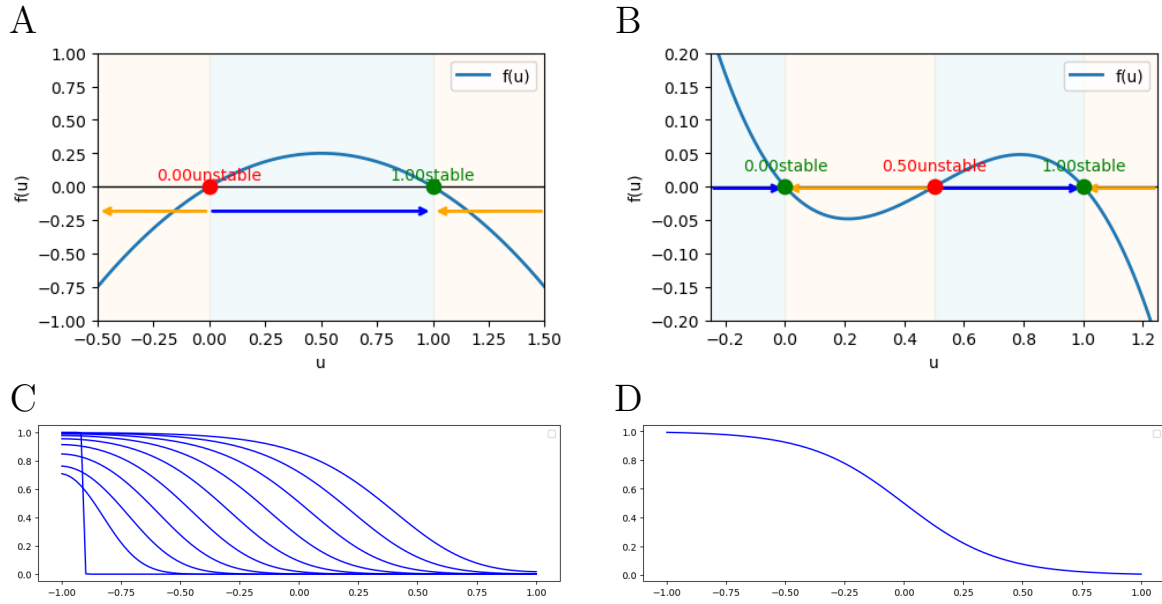


Figure S.3: Illustration of the interaction functions and travelling wave solutions in the Fisher and Allen–Cahn models. A. Interaction function in the Fisher model. The equilibrium $(0, 0)$ is unstable, while $(1, 0)$ is stable. B. Interaction function in the Allen–Cahn model, where $(0, 0)$ and $(1, 0)$ are stable equilibria and $(1/2, 0)$ is unstable. C. Numerical travelling wave solution of the Fisher model. Starting from a square-wave initial condition, the profile evolves into a logistic-like front propagating to the right with approximately constant speed. D. Travelling wave solution of the Allen–Cahn model. When $a = 1/2$, the wave speed satisfies $c = 0$, corresponding to a stationary solitary profile.

We now study the travelling wave equation on the phase plane. Define

$$p = \frac{du}{dz}.$$

Then the second-order ODE can be rewritten as

$$\begin{cases} \dot{u} = p, \\ \dot{p} = -cp - u(1 - u), \end{cases}$$

where the dot denotes differentiation with respect to z . The equilibria are

$$(u^*, p^*) = (0, 0), \quad (1, 0).$$

The Jacobian matrix is

$$\mathbf{J} = \begin{pmatrix} 0 & 1 \\ 2u - 1 & -c \end{pmatrix}.$$

At $(0, 0)$,

$$\mathbf{J}|_{(0,0)} = \begin{pmatrix} 0 & 1 \\ -1 & -c \end{pmatrix},$$

whose characteristic equation is

$$\lambda^2 + c\lambda + 1 = 0.$$

For $c \geq 2$, both eigenvalues are real and negative, so $(0, 0)$ is a stable node. At $(1, 0)$,

$$\mathbf{J}|_{(1,0)} = \begin{pmatrix} 0 & 1 \\ 1 & -c \end{pmatrix},$$

whose characteristic equation is

$$\lambda^2 + c\lambda - 1 = 0.$$

The eigenvalues are

$$\lambda = \frac{-c \pm \sqrt{c^2 + 4}}{2},$$

which have opposite signs. Therefore, $(1, 0)$ is a saddle point.

The travelling wave corresponds to a heteroclinic trajectory connecting $(1, 0)$ to $(0, 0)$ in the phase plane (Figure S.4). Let this trajectory be written as

$$p = \varphi(u).$$

Since

$$\frac{du}{dz} = \varphi(u),$$

we obtain

$$z - z_0 = \int_{u_0}^u \frac{1}{\varphi(u)} du.$$

Thus the travelling wave profile $u(z)$ can be reconstructed from the phase trajectory.

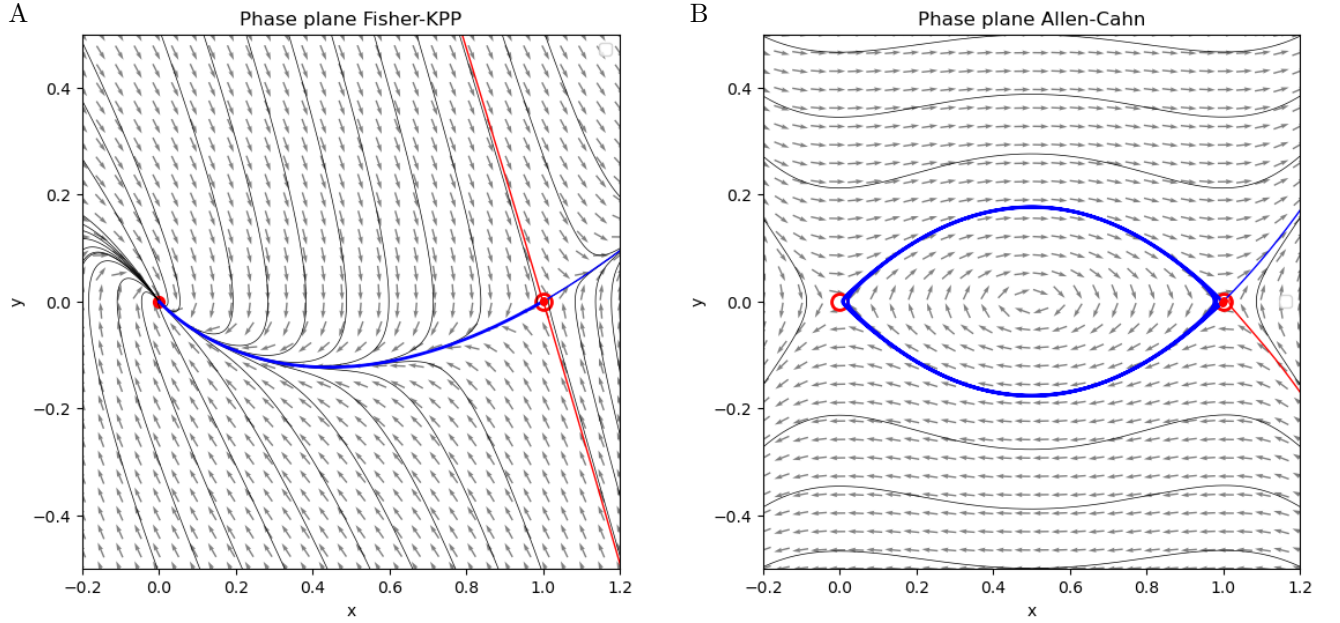


Figure S.4: Phase-plane trajectory corresponding to the travelling wave solution. A. The heteroclinic orbit connects the saddle point $(1, 0)$ to the stable node $(0, 0)$. B. The heteroclinic orbit connects stable point $(1, 0)$ to $(0, 0)$, forming a loop.

Consequently, the travelling wave solution satisfies the boundary conditions

$$\lim_{z \rightarrow -\infty} u(z) = 1, \quad \lim_{z \rightarrow +\infty} u(z) = 0.$$

To approximate the wave profile, we consider the large-speed limit $c \rightarrow \infty$. Introduce the stretched coordinate

$$\xi = \frac{z}{c}.$$

The travelling wave equation becomes

$$\frac{1}{c^2} \frac{d^2 u}{d\xi^2} + \frac{du}{d\xi} + u(1 - u) = 0.$$

Assume an asymptotic expansion of the form

$$u(\xi) = u_0(\xi) + \frac{1}{c^2}u_1(\xi) + \mathcal{O}\left(\frac{1}{c^4}\right).$$

Collecting terms of equal order in c^{-1} gives

$$\frac{du_0}{d\xi} + u_0(1 - u_0) = 0,$$

and

$$\frac{d^2u_0}{d\xi^2} + \frac{du_1}{d\xi} + u_1(1 - 2u_0) = 0.$$

Imposing the symmetry condition

$$u(0) = \frac{1}{2},$$

we obtain

$$u_0(z) = \frac{1}{1 + e^{z/c}},$$

and

$$u_1(z) = \frac{e^{z/c}}{(1 + e^{z/c})^2} \ln \left(\frac{4e^{z/c}}{(1 + e^{z/c})^2} \right).$$

Therefore,

$$u(z) \approx u_0(z) + \frac{1}{c^2}u_1(z).$$

This approximation remains accurate even for the minimal wave speed $c = 2$.

C Allen–Cahn Equation

Here we consider the Allen–Cahn equation (Allen and Cahn, 1975)

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + u(1 - u)(u - a).$$

Assuming a travelling wave solution

$$u(x, t) = u(z), \quad z = x - ct,$$

we obtain the ordinary differential equation

$$\frac{d^2u}{dz^2} + c \frac{du}{dz} + u(1 - u)(u - a) = 0.$$

Introducing

$$p = \frac{du}{dz},$$

the equation can be rewritten as the first-order system

$$\begin{cases} \dot{u} = p, \\ \dot{p} = -cp - u(1-u)(u-a), \end{cases}$$

where the dot denotes differentiation with respect to z .

The Jacobian matrix is

$$\mathbf{J} = \begin{pmatrix} 0 & 1 \\ 3u^2 - 2(1+a)u + a & -c \end{pmatrix}.$$

The equilibria (Figure S.3B) are

$$(u^*, p^*) = (0, 0), \quad (1, 0), \quad (a, 0).$$

At $(0, 0)$,

$$\mathbf{J}|_{(0,0)} = \begin{pmatrix} 0 & 1 \\ a & -c \end{pmatrix}.$$

Since

$$\text{tr}(\mathbf{J}) = -c < 0, \quad \det(\mathbf{J}) = -a < 0,$$

the equilibrium $(0, 0)$ is a saddle point.

At $(1, 0)$,

$$\mathbf{J}|_{(1,0)} = \begin{pmatrix} 0 & 1 \\ 1-a & -c \end{pmatrix}.$$

Similarly,

$$\text{tr}(\mathbf{J}) = -c < 0, \quad \det(\mathbf{J}) = -(1-a) < 0,$$

so $(1, 0)$ is also a saddle point.

At $(a, 0)$,

$$\mathbf{J}|_{(a,0)} = \begin{pmatrix} 0 & 1 \\ a(a-1) & -c \end{pmatrix}.$$

In this case,

$$\text{tr}(\mathbf{J}) = -c < 0, \quad \det(\mathbf{J}) = a(1-a) > 0.$$

Therefore, $(a, 0)$ is stable. Its characteristic equation is

$$\lambda^2 + c\lambda + a(1 - a) = 0,$$

with eigenvalues

$$\lambda = \frac{-c \pm \sqrt{c^2 - 4a(1 - a)}}{2}.$$

Depending on the sign of

$$\Delta = c^2 - 4a(1 - a),$$

the equilibrium may be either a stable node or a spiral sink.

The travelling wave solution corresponds to a heteroclinic trajectory connecting the two saddle points $(1, 0)$ and $(0, 0)$ (Figure S.4). Such a connection only exists for a particular wave speed c .

Symmetric Case: $a = \frac{1}{2}$

When

$$a = \frac{1}{2},$$

the system becomes symmetric with respect to the two stable states $u = 0$ and $u = 1$. In this case, the travelling wave speed satisfies

$$c = 0,$$

and the travelling wave becomes a stationary interface.

The equation reduces to

$$\frac{d^2u}{dz^2} = u(1 - u) \left(\frac{1}{2} - u \right).$$

Multiplying both sides by $\frac{du}{dz}$ gives

$$\begin{aligned} \frac{d^2u}{dz^2} \frac{du}{dz} &= u(1 - u) \left(\frac{1}{2} - u \right) \frac{du}{dz}, \\ \frac{1}{2} \frac{d}{dz} \left(\frac{du}{dz} \right)^2 &= u(1 - u) \left(\frac{1}{2} - u \right) \frac{du}{dz}. \end{aligned}$$

Integrating with respect to z yields

$$\frac{1}{2} \left(\frac{du}{dz} \right)^2 = \frac{1}{4} u^2 (1 - u)^2.$$

Hence,

$$\frac{du}{dz} = \frac{\sqrt{2}}{2} u(1-u).$$

Separating variables,

$$\begin{aligned} \frac{1}{u(1-u)} du &= \frac{\sqrt{2}}{2} dz, \\ \left(\frac{1}{u} + \frac{1}{1-u} \right) du &= \frac{\sqrt{2}}{2} dz. \end{aligned}$$

After integration,

$$\ln u - \ln(1-u) = \frac{z}{\sqrt{2}} + C.$$

Therefore,

$$u(z) = \frac{1}{1 + e^{z/\sqrt{2} + C}}.$$

Choosing the interface center such that

$$u(0) = \frac{1}{2},$$

we obtain $C = 0$, and thus (Figure S.3D)

$$u(z) = \frac{1}{1 + e^{z/\sqrt{2}}}.$$

Energy Interpretation

The travelling wave equation can also be studied using an energy method. Consider

$$\frac{d^2u}{dz^2} + c \frac{du}{dz} + u(1-u)(u-a) = 0.$$

Multiplying both sides by $\frac{du}{dz}$ gives

$$\begin{aligned} \left(\frac{d^2u}{dz^2} + c \frac{du}{dz} + u(1-u)(u-a) \right) \frac{du}{dz} &= 0, \\ \frac{1}{2} \frac{d}{dz} \left(\frac{du}{dz} \right)^2 + c \left(\frac{du}{dz} \right)^2 + u(1-u)(u-a) \frac{du}{dz} &= 0. \end{aligned}$$

Define the potential function $W(u)$ by

$$\frac{dW}{du} = u(1-u)(u-a).$$

Integrating,

$$W(u) = -\frac{1}{4}u^4 + \frac{1+a}{3}u^3 - \frac{a}{2}u^2,$$

where we choose $W(0) = 0$. Then

$$\frac{dW}{dz} = \frac{dW}{du} \frac{du}{dz}.$$

Substituting into the previous equation yields

$$\frac{d}{dz} \left[\frac{1}{2} \left(\frac{du}{dz} \right)^2 + W(u) \right] = -c \left(\frac{du}{dz} \right)^2 \leq 0.$$

Therefore, the quantity

$$E(u) = \frac{1}{2} \left(\frac{du}{dz} \right)^2 + W(u)$$

acts as a decreasing energy functional. Integrating over the entire domain gives

$$\left[\frac{1}{2} \left(\frac{du}{dz} \right)^2 + W(u) \right] \Big|_{-\infty}^{+\infty} = -c \int_{-\infty}^{+\infty} \left(\frac{du}{dz} \right)^2 dz.$$

Using the boundary conditions

$$\lim_{z \rightarrow -\infty} u(z) = 1, \quad \lim_{z \rightarrow +\infty} u(z) = 0,$$

and assuming

$$u'(z) \rightarrow 0 \quad (z \rightarrow \pm\infty),$$

we obtain

$$W(1) - W(0) = -c \int_{-\infty}^{+\infty} \left(\frac{du}{dz} \right)^2 dz.$$

Hence the wave speed is

$$c = \frac{W(0) - W(1)}{\int_{-\infty}^{+\infty} \left(\frac{du}{dz} \right)^2 dz}.$$

For

$$a = \frac{1}{2},$$

we have

$$W(0) = W(1),$$

which immediately implies

$$c = 0.$$

General Wave Speed

Motivated by the stationary solution, assume the travelling wave profile satisfies

$$\frac{du}{dz} = ku(1 - u),$$

for some constant k . Then

$$\begin{aligned} \frac{d^2u}{dz^2} &= k(1 - 2u)\frac{du}{dz}, \\ &= k^2u(1 - u)(1 - 2u). \end{aligned}$$

Substituting into the travelling wave equation gives

$$k^2u(1 - u)(1 - 2u) + cku(1 - u) + u(1 - u)(u - a) = 0.$$

Factoring out $u(1 - u)$,

$$u(1 - u) [k^2(1 - 2u) + ck + u - a] = 0.$$

Matching coefficients of u and constants yields

$$2k^2 = 1, \quad k^2 + ck - a = 0.$$

Thus,

$$k = \frac{\sqrt{2}}{2},$$

and

$$c = \sqrt{2} \left(a - \frac{1}{2} \right).$$

Therefore, the Allen–Cahn equation admits an exact travelling wave solution with speed

$$c = \sqrt{2} \left(a - \frac{1}{2} \right),$$

while the wave profile retains the logistic form

$$u(z) = \frac{1}{1 + e^{z/\sqrt{2}}}.$$

D Appendix Figure

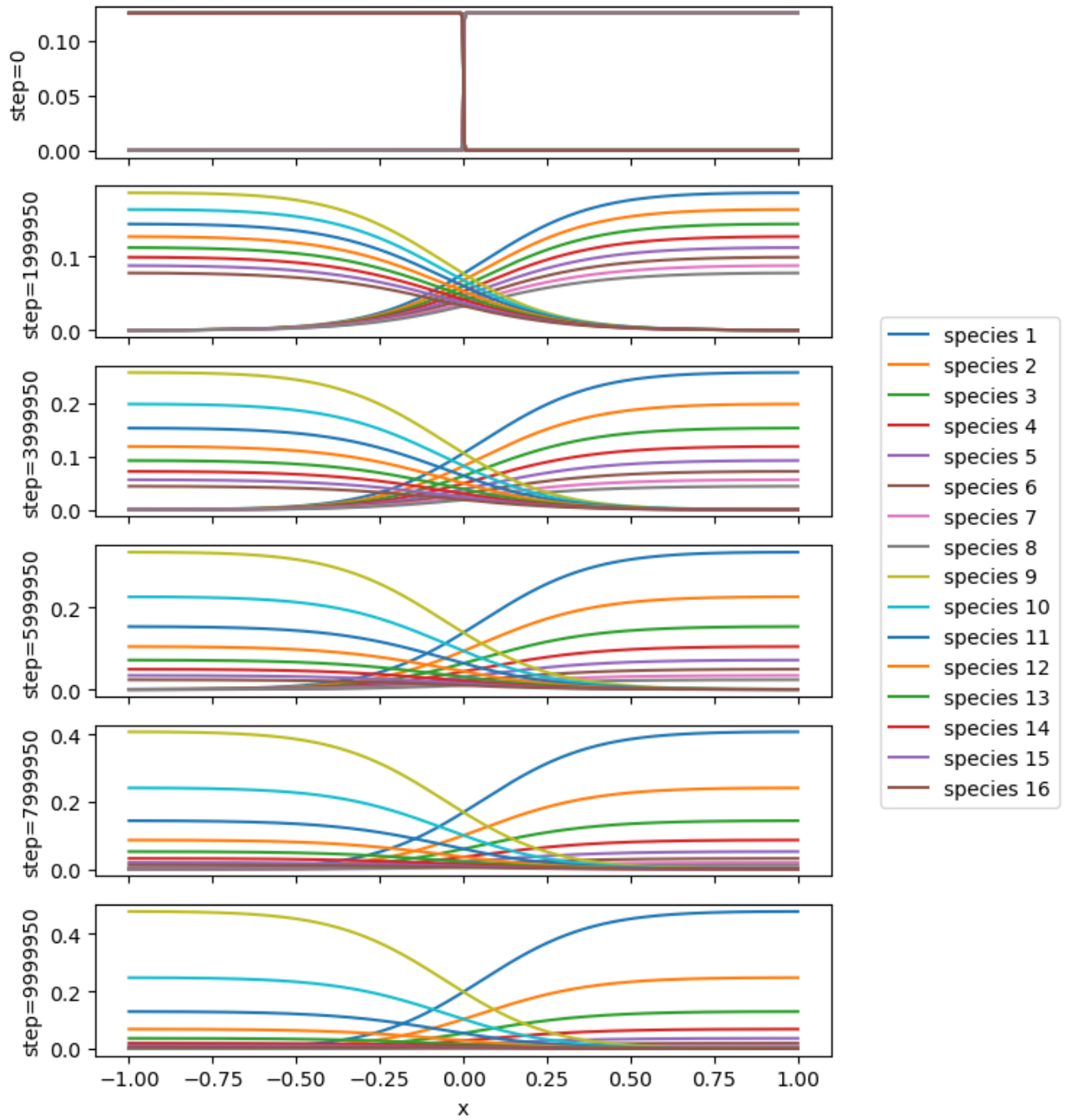


Figure S.5: More detailed simulation process of multispecies with different diffusion rate

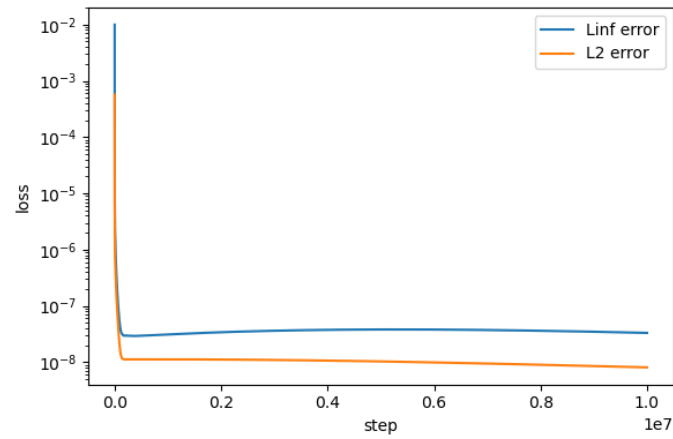


Figure S.6: The loss of multispecies with different diffusion rate

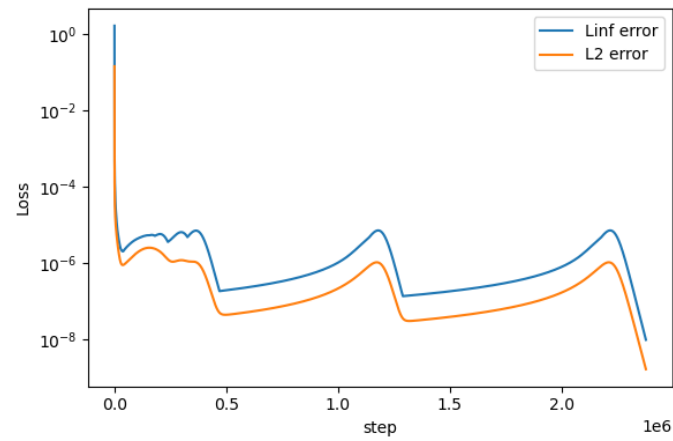


Figure S.7: The loss of two symmetric species competing in a larger scale.

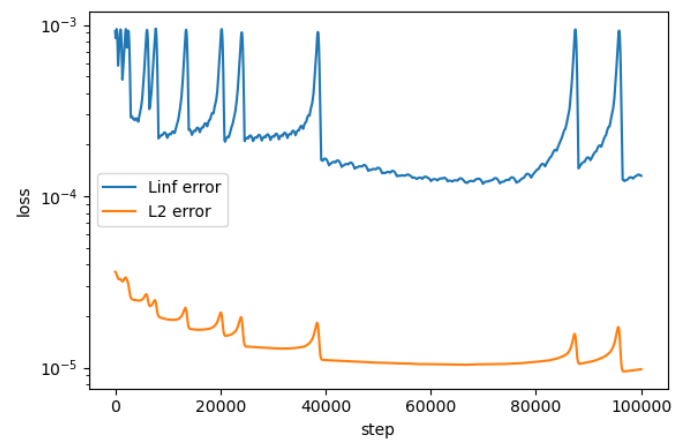


Figure S.8: The loss of 2D simulation

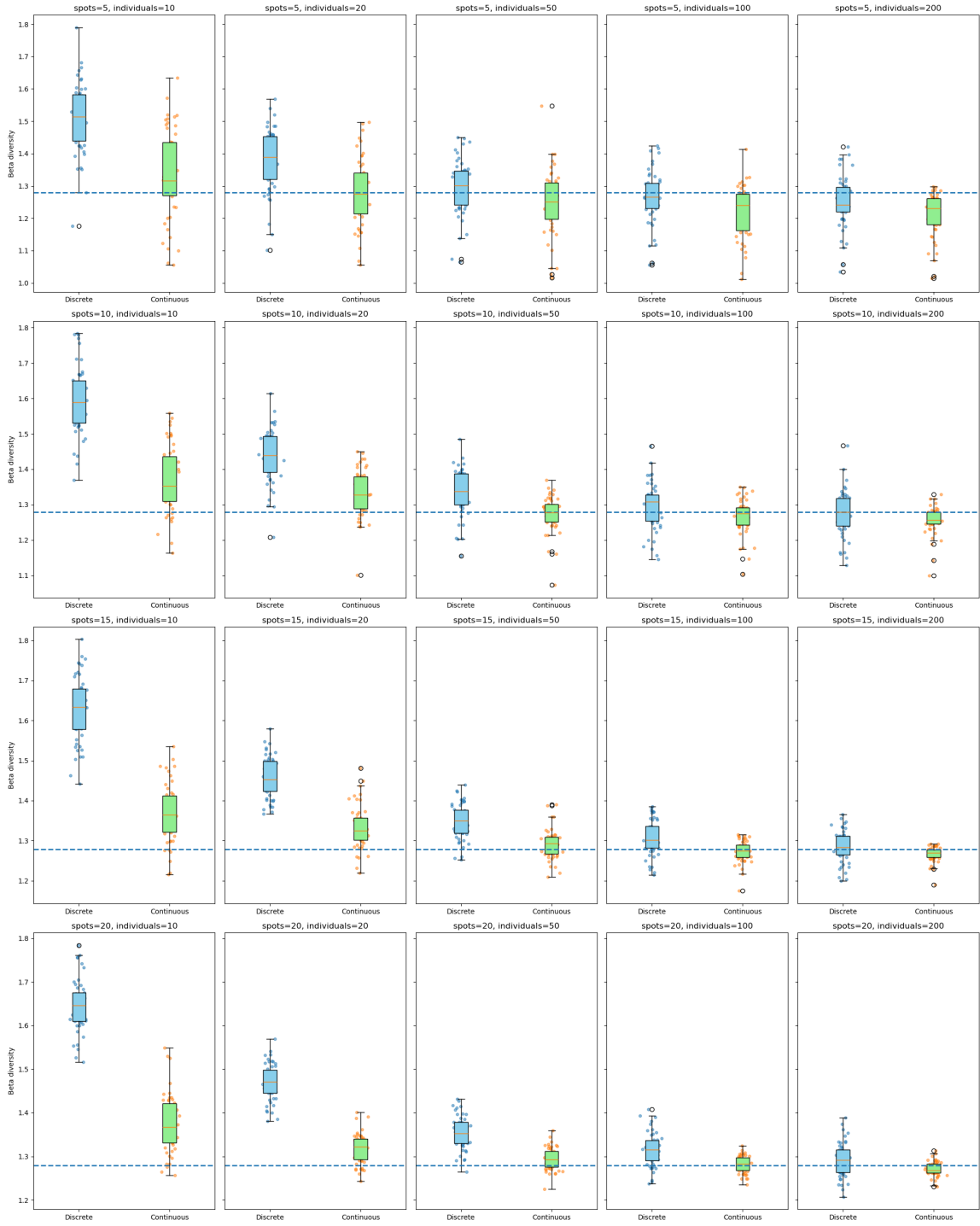


Figure S.9: Comparison between discrete and continuous β estimator. $k = 5, 10, 15, 20$, $n = 10, 20, 50, 100, 200$.

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